

# St. Martins Marsh Aquatic Preserve

## SEACAR Water Quality Analysis

Last compiled on 09 April, 2026

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# Indicators

## Nutrients

### Total Nitrogen - Discrete

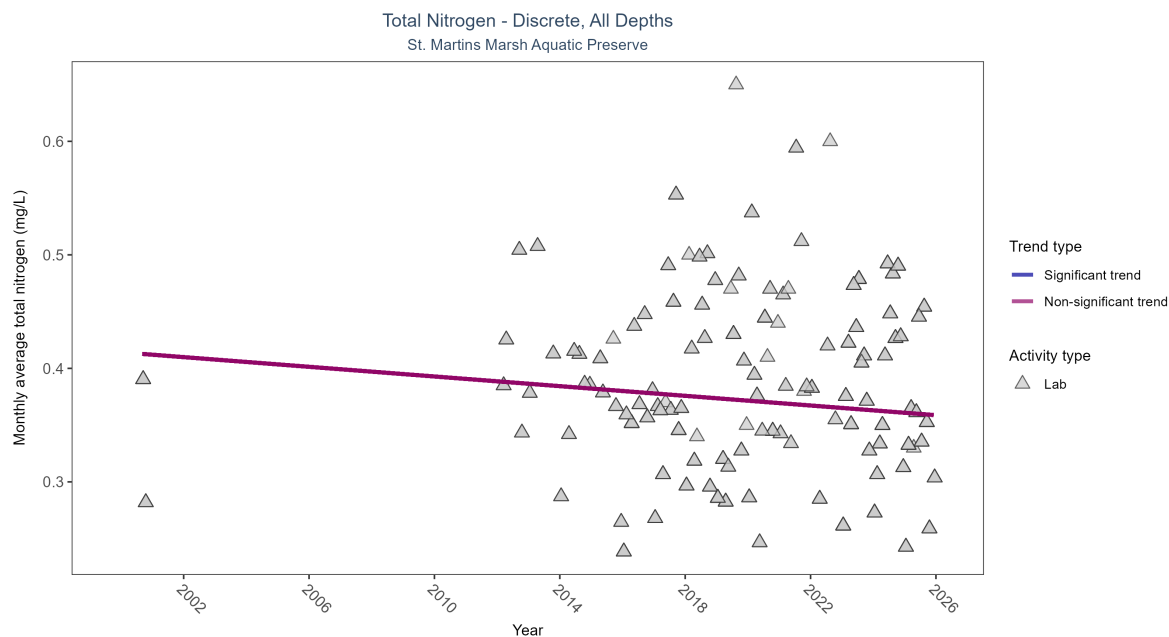


Figure 1: Scatter plot of monthly average total nitrogen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only nitrogen values obtained from laboratory analyses (triangles) are included in the plot.

Table 1: Seasonal Kendall-Tau Results for - Total Nitrogen

| Activity Type | Statistical Trend    | Sample Count | Years with Data | Period of Record | Median Result Value | Tau      | Sen Intercept | Sen Slope | P      |
|---------------|----------------------|--------------|-----------------|------------------|---------------------|----------|---------------|-----------|--------|
| Lab           | No significant trend | 673          | 15              | 2000 - 2025      | 0.365               | -0.08575 | 0.41416       | -0.00213  | 0.1966 |

Total nitrogen showed no detectable trend between 2000 and 2025.

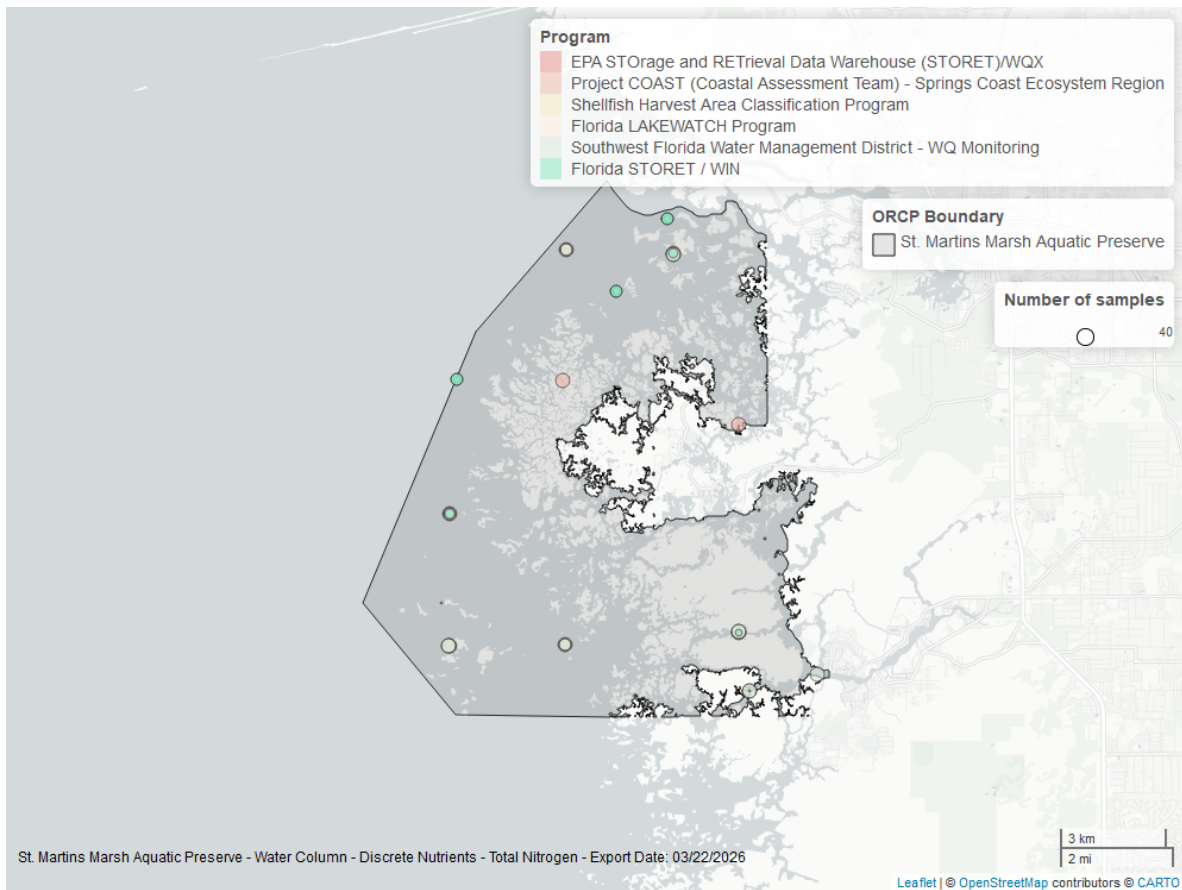


Figure 2: Map showing location of discrete water quality sampling locations within the boundaries of *St. Martins Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## Total Phosphorus - Discrete

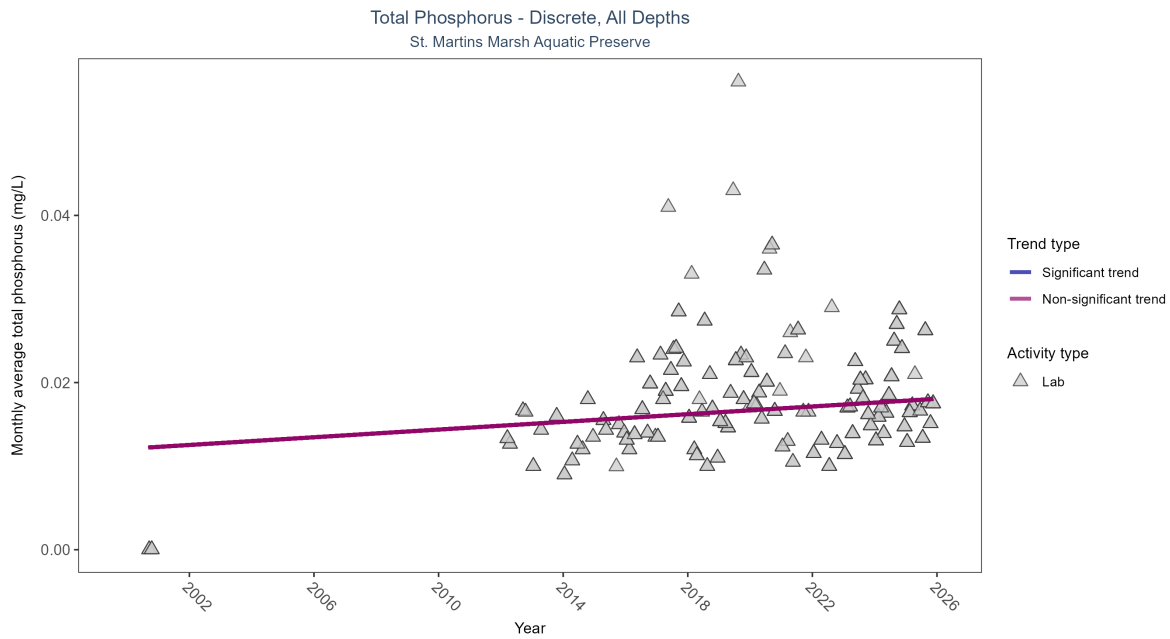


Figure 3: Scatter plot of monthly average total phosphorus over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only phosphorus values obtained from laboratory analyses (triangles) are included in the plot.

Table 2: Seasonal Kendall-Tau Results for - Total Phosphorus

| Activity Type | Statistical Trend    | Sample Count | Years with Data | Period of Record | Median Result Value | Tau    | Sen Intercept | Sen Slope | P      |
|---------------|----------------------|--------------|-----------------|------------------|---------------------|--------|---------------|-----------|--------|
| Lab           | No significant trend | 663          | 15              | 2000 - 2025      | 0.016               | 0.0941 | 0.01207       | 0.00023   | 0.1113 |

Total phosphorus showed no detectable trend between 2000 and 2025.



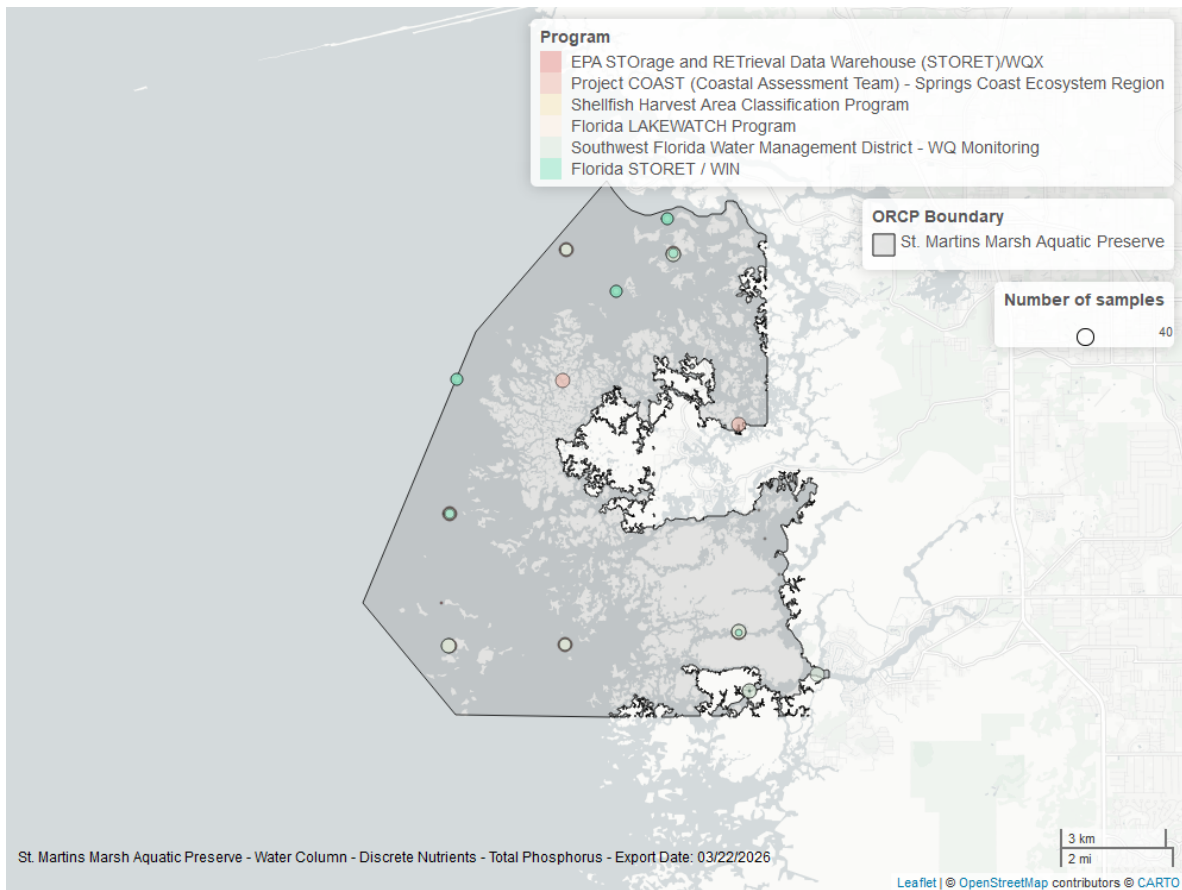


Figure 4: Map showing location of discrete water quality sampling locations within the boundaries of *St. Martins Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## Water Quality

### Dissolved Oxygen - Continuous

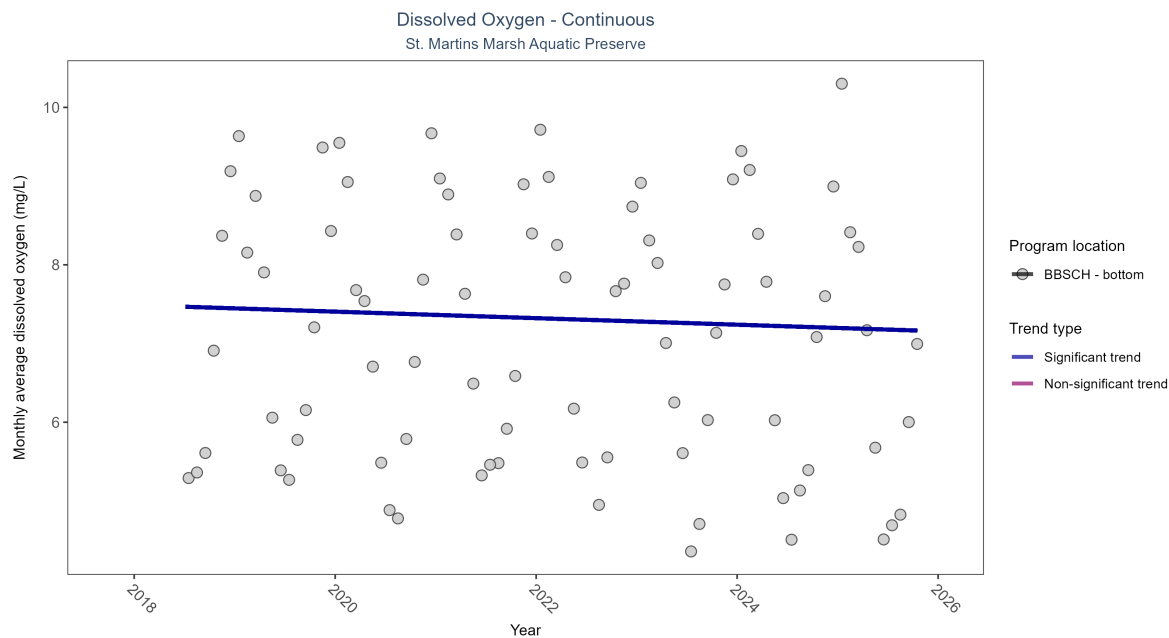


Figure 5: Scatter plot of monthly average dissolved oxygen over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 3: Seasonal Kendall-Tau Results - Dissolved Oxygen

| Program Location | Statistical Trend              | Sample Count | Years with Data | Period of Record | Median Result Value | Tau   | Sen Intercept | Sen Slope | P      |
|------------------|--------------------------------|--------------|-----------------|------------------|---------------------|-------|---------------|-----------|--------|
| BBSCH            | Significantly decreasing trend | 227936       | 8               | 2018 - 2025      | 7.2                 | -0.21 | 7.49          | -0.04     | 0.0217 |

At one program location, monthly average dissolved oxygen decreased by 0.04 mg/L per year.

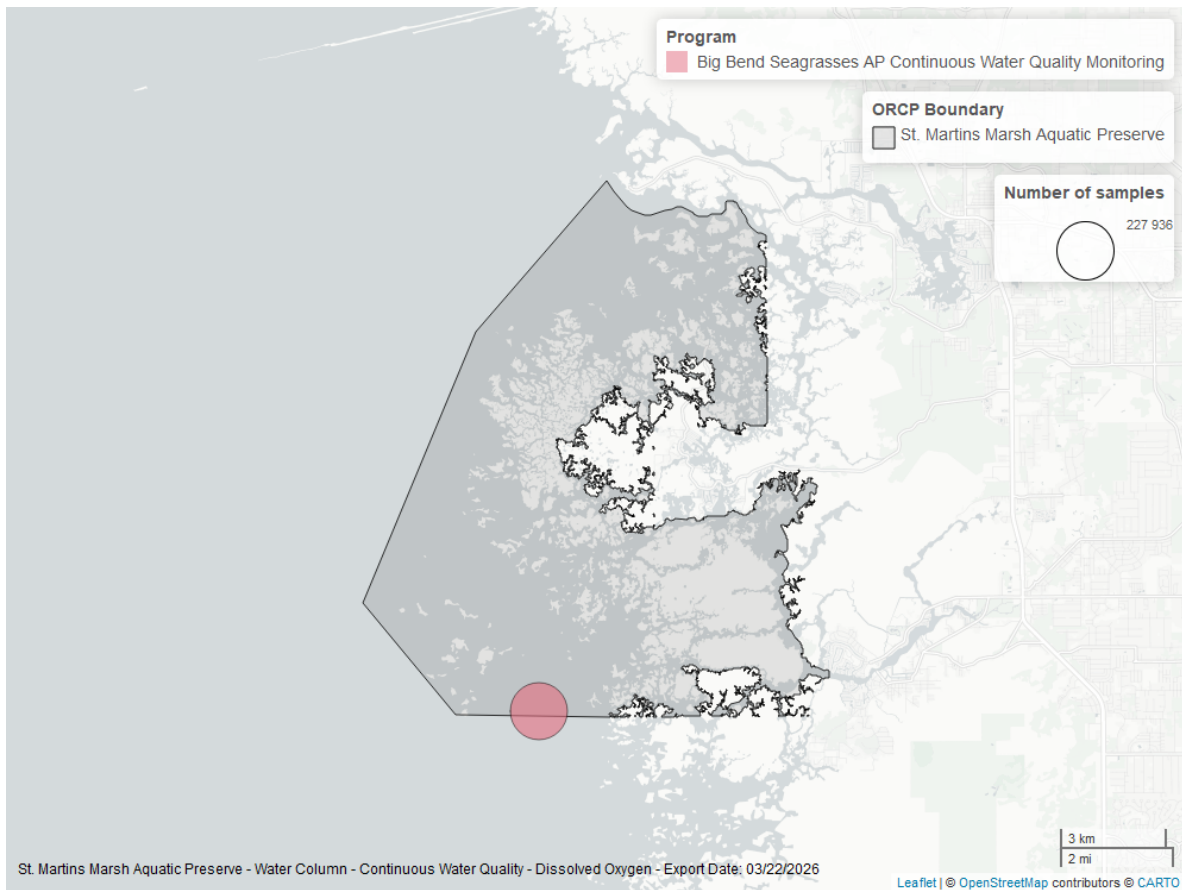


Figure 6: Map showing location of dissolved oxygen continuous water quality sampling locations within the boundaries of *St. Martins Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## Dissolved Oxygen - Discrete

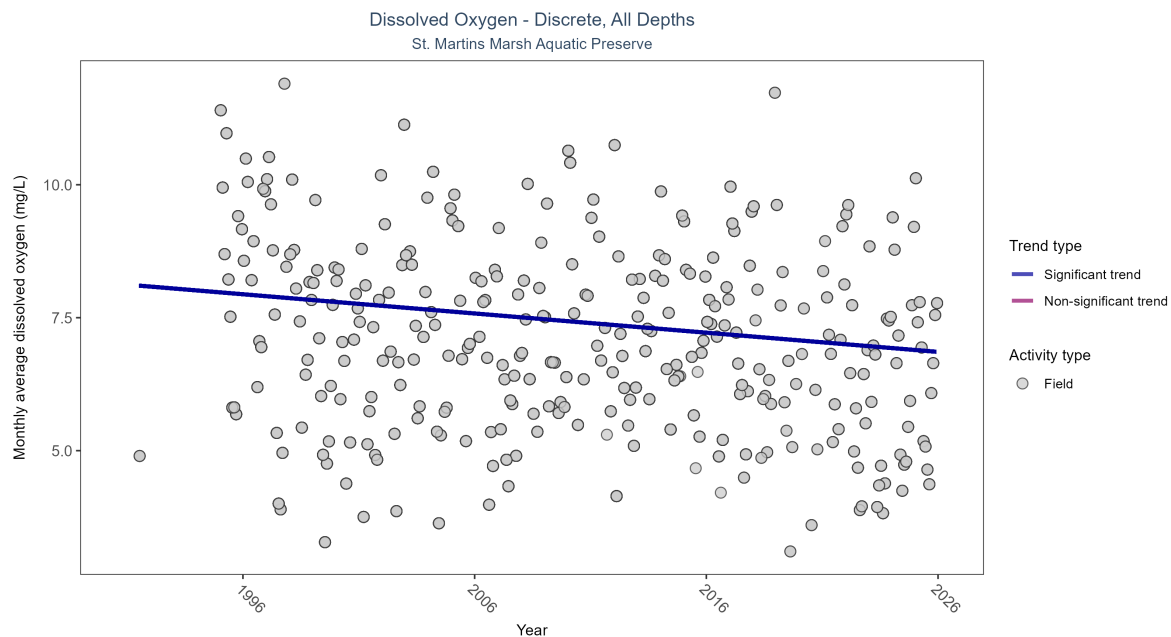


Figure 7: Scatter plot of monthly average dissolved oxygen over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen values measured in the field (circles) are included in the plot.

Table 4: Seasonal Kendall-Tau Results for - Dissolved Oxygen

| Activity Type | Statistical Trend              | Sample Count | Years with Data | Period of Record | Median Result Value | Tau      | Sen Intercept | Sen Slope | P |
|---------------|--------------------------------|--------------|-----------------|------------------|---------------------|----------|---------------|-----------|---|
| Field         | Significantly decreasing trend | 7885         | 32              | 1991 - 2025      | 7.3                 | -0.19439 | 8.12148       | -0.03613  | 0 |

Monthly average dissolved oxygen decreased by 0.04 mg/L per year.

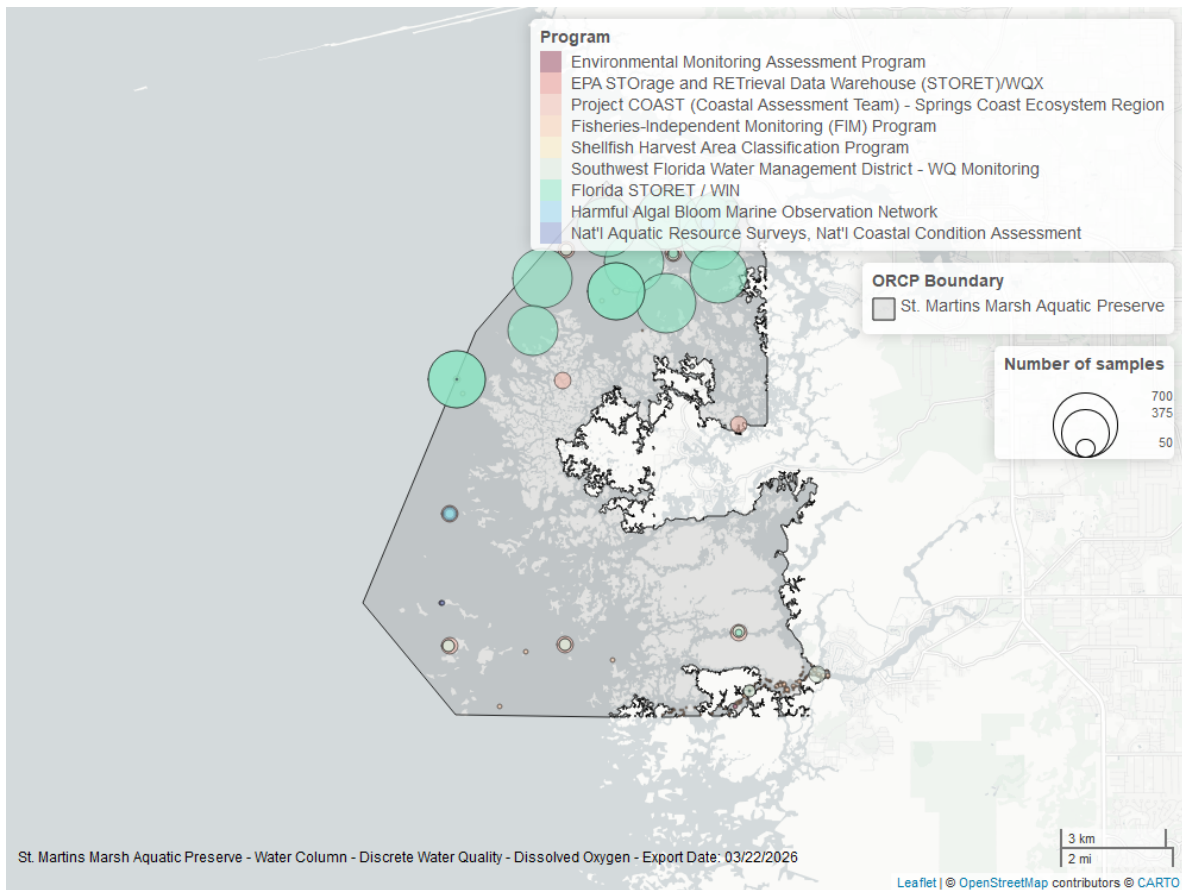


Figure 8: Map showing location of discrete water quality sampling locations within the boundaries of *St. Martins Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## Dissolved Oxygen Saturation - Continuous

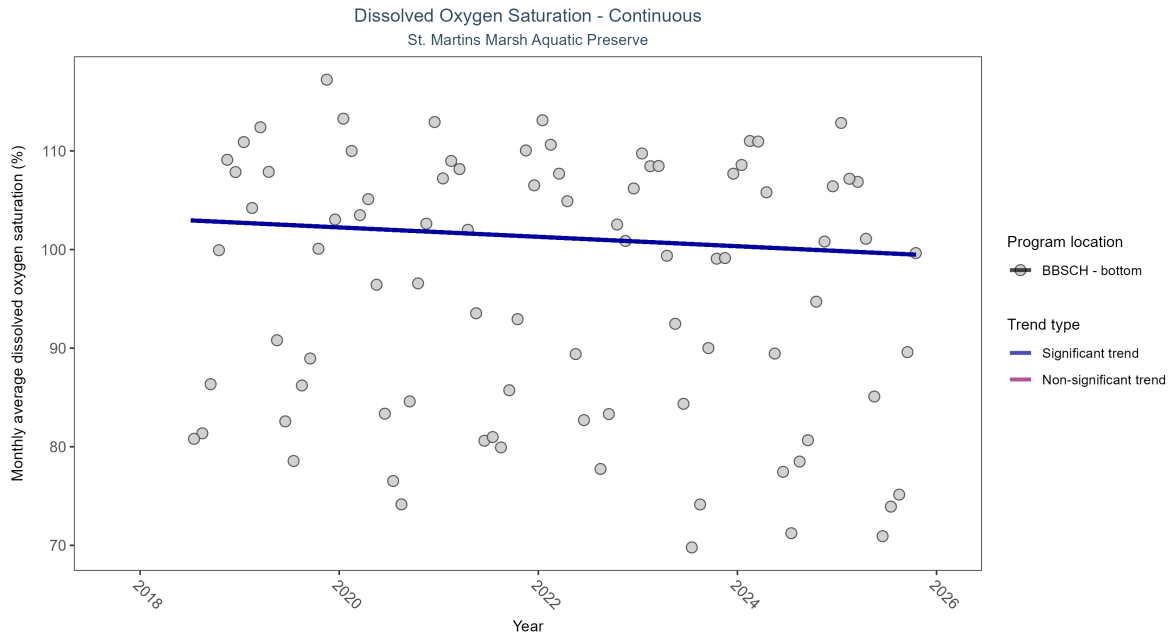


Figure 9: Scatter plot of monthly average dissolved oxygen saturation over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 5: Seasonal Kendall-Tau Results - Dissolved Oxygen Saturation

| Program Location | Statistical Trend              | Sample Count | Years with Data | Period of Record | Median Result Value | Tau   | Sen Intercept | Sen Slope | P      |
|------------------|--------------------------------|--------------|-----------------|------------------|---------------------|-------|---------------|-----------|--------|
| BBSCH            | Significantly decreasing trend | 227936       | 8               | 2018 - 2025      | 97.7                | -0.26 | 103.2         | -0.48     | 0.0041 |

At one program location, monthly average dissolved oxygen saturation decreased by 0.48% per year.

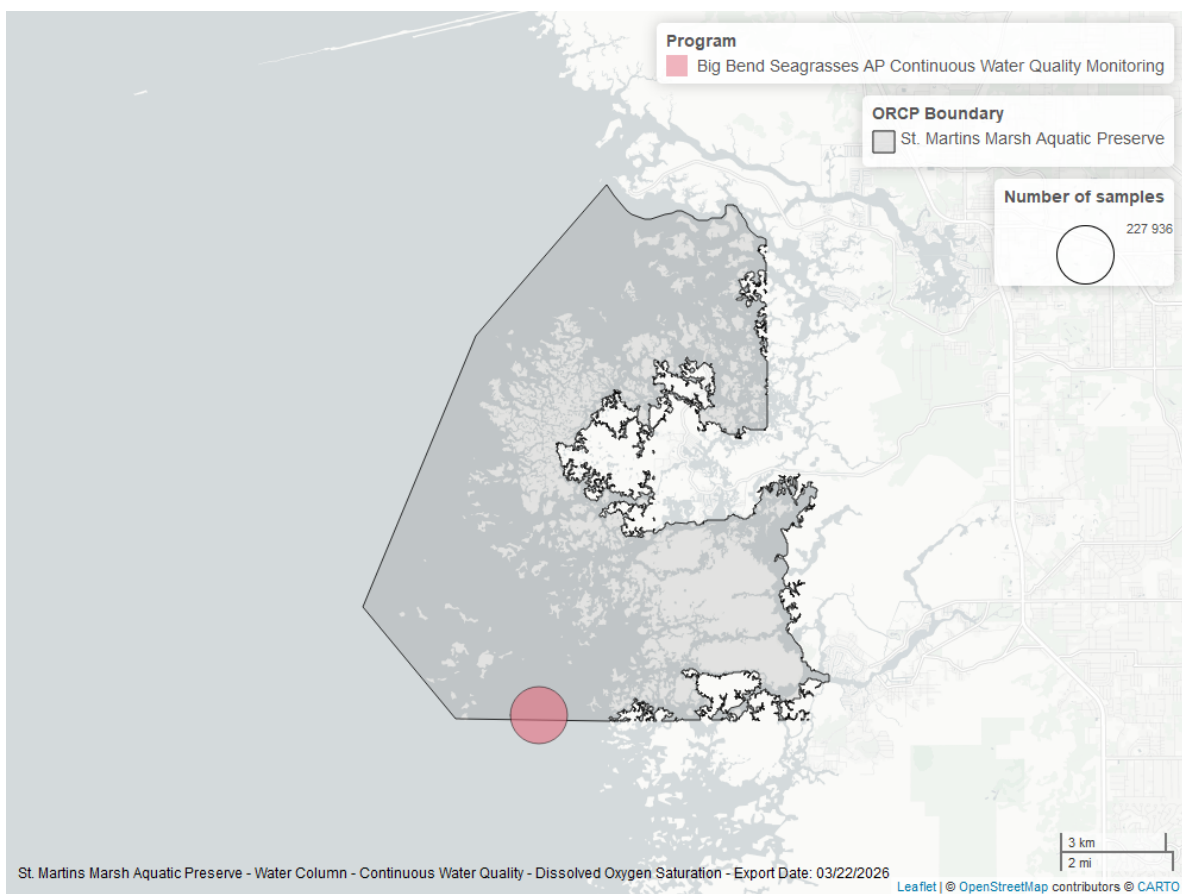


Figure 10: Map showing location of dissolved oxygen saturation continuous water quality sampling locations within the boundaries of *St. Martins Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## Dissolved Oxygen Saturation - Discrete

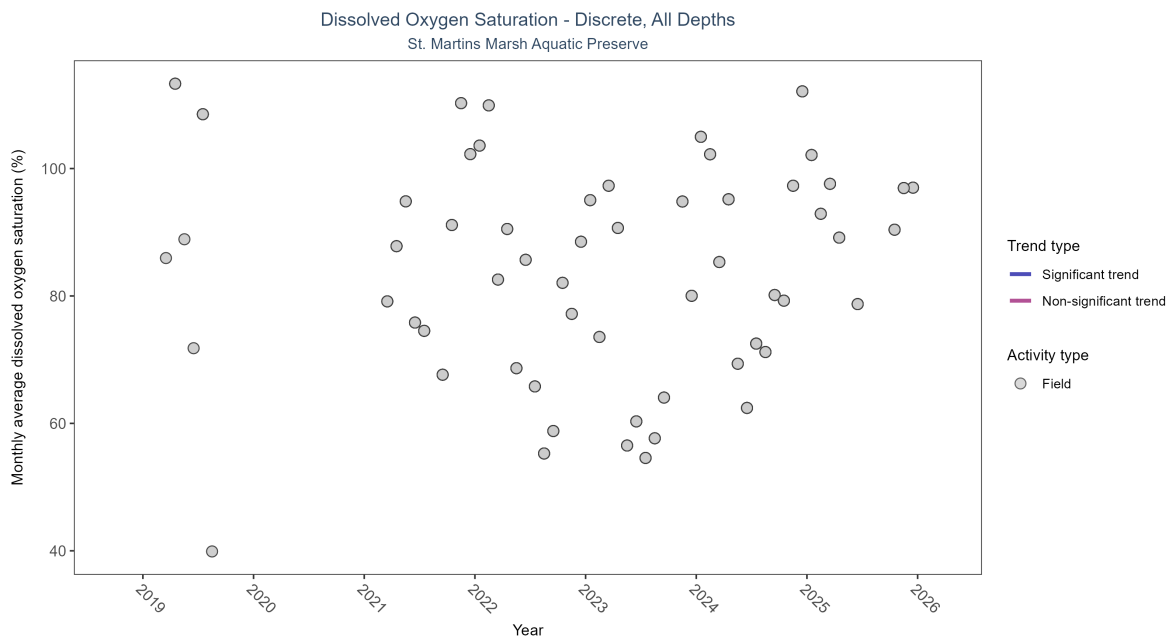


Figure 11: Scatter plot of monthly average dissolved oxygen saturation over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only dissolved oxygen saturation values measured in the field (circles) are included in the plot.

Table 6: Seasonal Kendall-Tau Results for - Dissolved Oxygen Saturation

| Activity Type | Statistical Trend                    | Sample Count | Years with Data | Period of Record | Median Result Value | Tau | Sen Intercept | Sen Slope | P |
|---------------|--------------------------------------|--------------|-----------------|------------------|---------------------|-----|---------------|-----------|---|
| Field         | Insufficient data to calculate trend | 461          | 6               | 2019 - 2025      | 86.6                | -   | -             | -         | - |

There was insufficient data to fit a model for dissolved oxygen saturation.



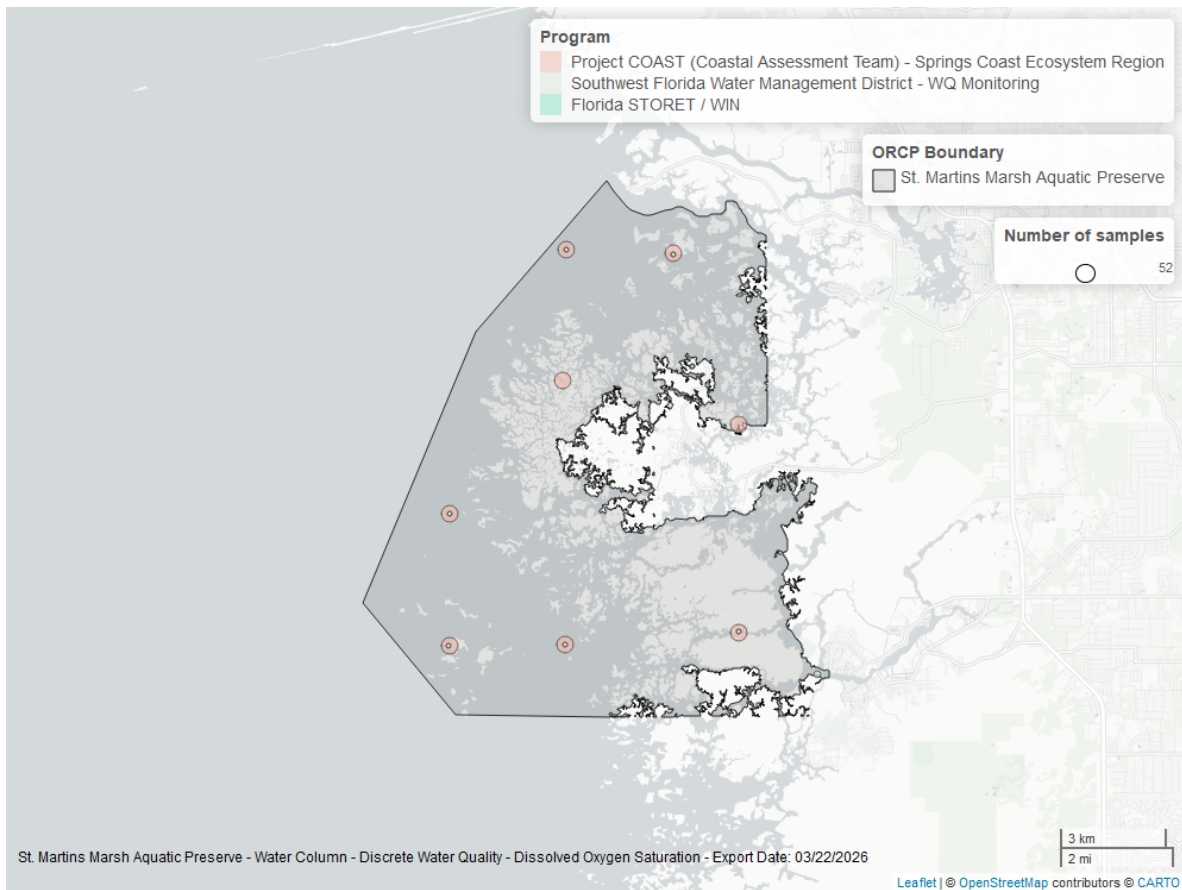


Figure 12: Map showing location of discrete water quality sampling locations within the boundaries of *St. Martins Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## Salinity - Discrete

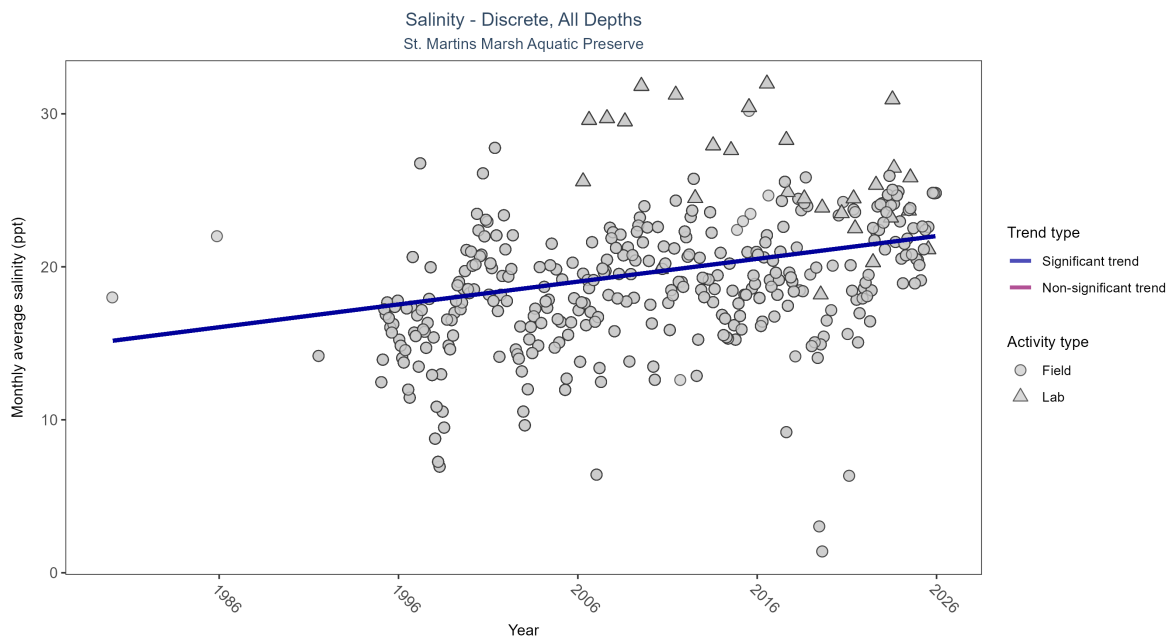


Figure 13: Scatter plot of monthly average salinity over time. If the time series included ten or more years of discrete observations, significant (blue) or non-significant (magenta) trend lines are also shown. Discrete salinity values derived from grab samples analyzed in the field (circles) or the laboratory (triangles) are both included in the plot.

Table 7: Seasonal Kendall-Tau Results for - Salinity

| Activity Type | Statistical Trend              | Sample Count | Years with Data | Period of Record | Median Result Value | Tau     | Sen Intercept | Sen Slope | P |
|---------------|--------------------------------|--------------|-----------------|------------------|---------------------|---------|---------------|-----------|---|
| All           | Significantly increasing trend | 9666         | 34              | 1980 - 2025      | 18.4                | 0.25999 | 15.158        | 0.14896   | 0 |

Monthly average salinity increased by 0.15 ppt per year.

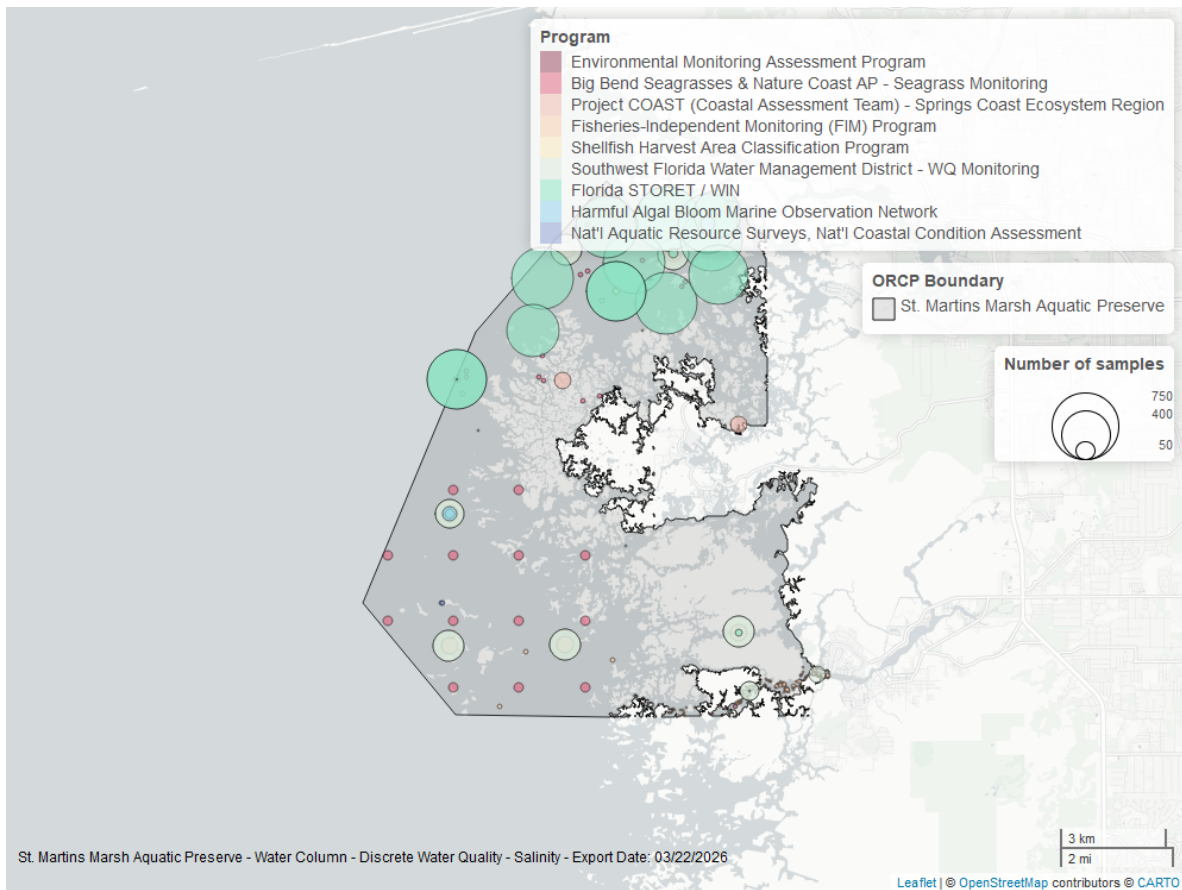


Figure 14: Map showing location of discrete water quality sampling locations within the boundaries of *St. Martins Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## Salinity - Continuous

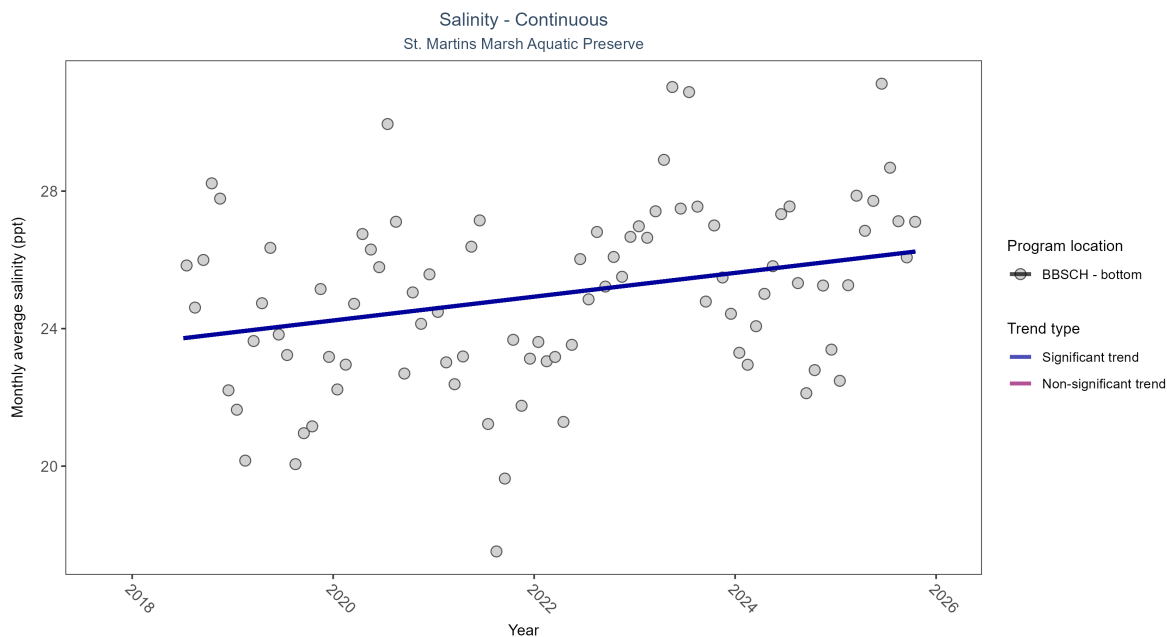


Figure 15: Scatter plot of monthly average salinity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 8: Seasonal Kendall-Tau Results - Salinity

| Program Location | Statistical Trend              | Sample Count | Years with Data | Period of Record | Median Result Value | Tau  | Sen Intercept | Sen Slope | P      |
|------------------|--------------------------------|--------------|-----------------|------------------|---------------------|------|---------------|-----------|--------|
| BBSCH            | Significantly increasing trend | 235777       | 8               | 2018 - 2025      | 25.2                | 0.29 | 23.55         | 0.35      | 0.0015 |

At one program location, monthly average salinity increased by 0.35 ppt per year.

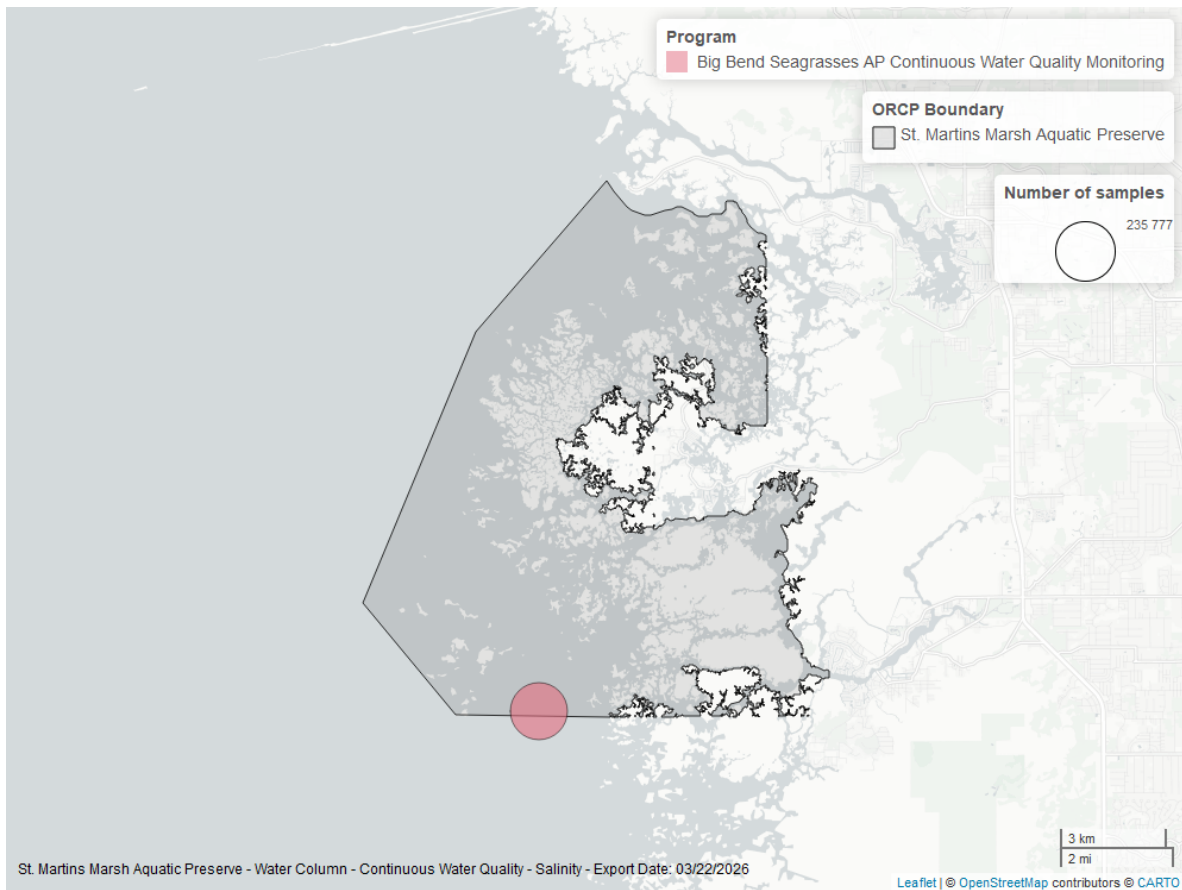


Figure 16: Map showing location of salinity continuous water quality sampling locations within the boundaries of *St. Martins Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## Water Temperature - Continuous

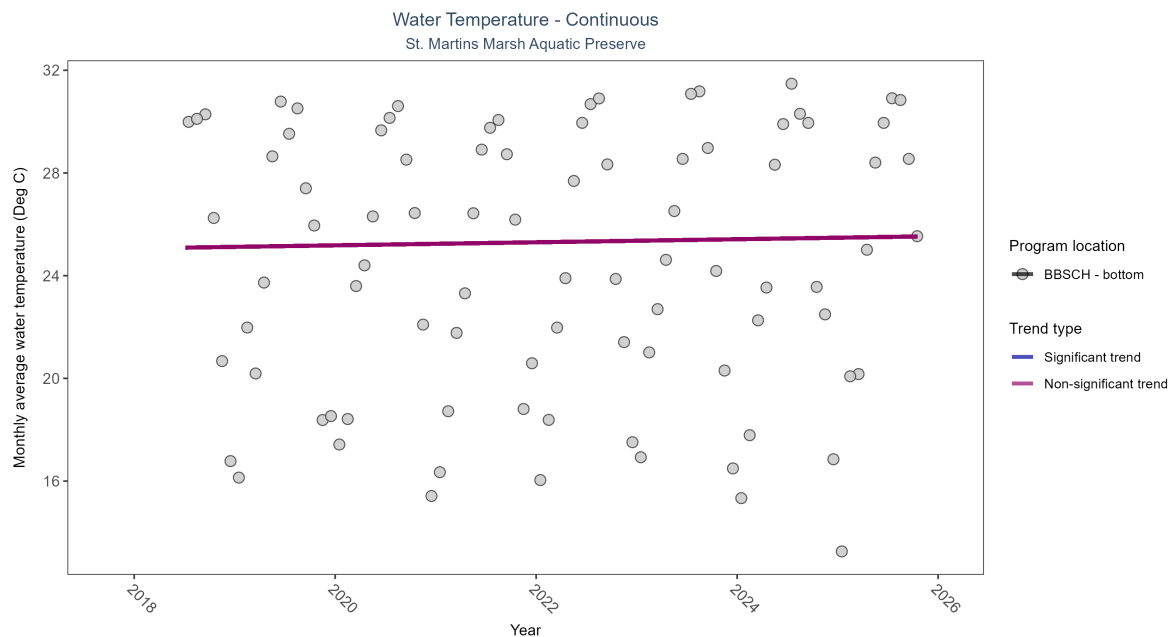


Figure 17: Scatter plot of monthly average water temperature over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 9: Seasonal Kendall-Tau Results - Water Temperature

| Program Location | Statistical Trend    | Sample Count | Years with Data | Period of Record | Median Result Value | Tau  | Sen Intercept | Sen Slope | P      |
|------------------|----------------------|--------------|-----------------|------------------|---------------------|------|---------------|-----------|--------|
| BBSCH            | No significant trend | 237837       | 8               | 2018 - 2025      | 25.3                | 0.06 | 25.06         | 0.06      | 0.4934 |

No detectable change in monthly average water temperature was observed at one location.

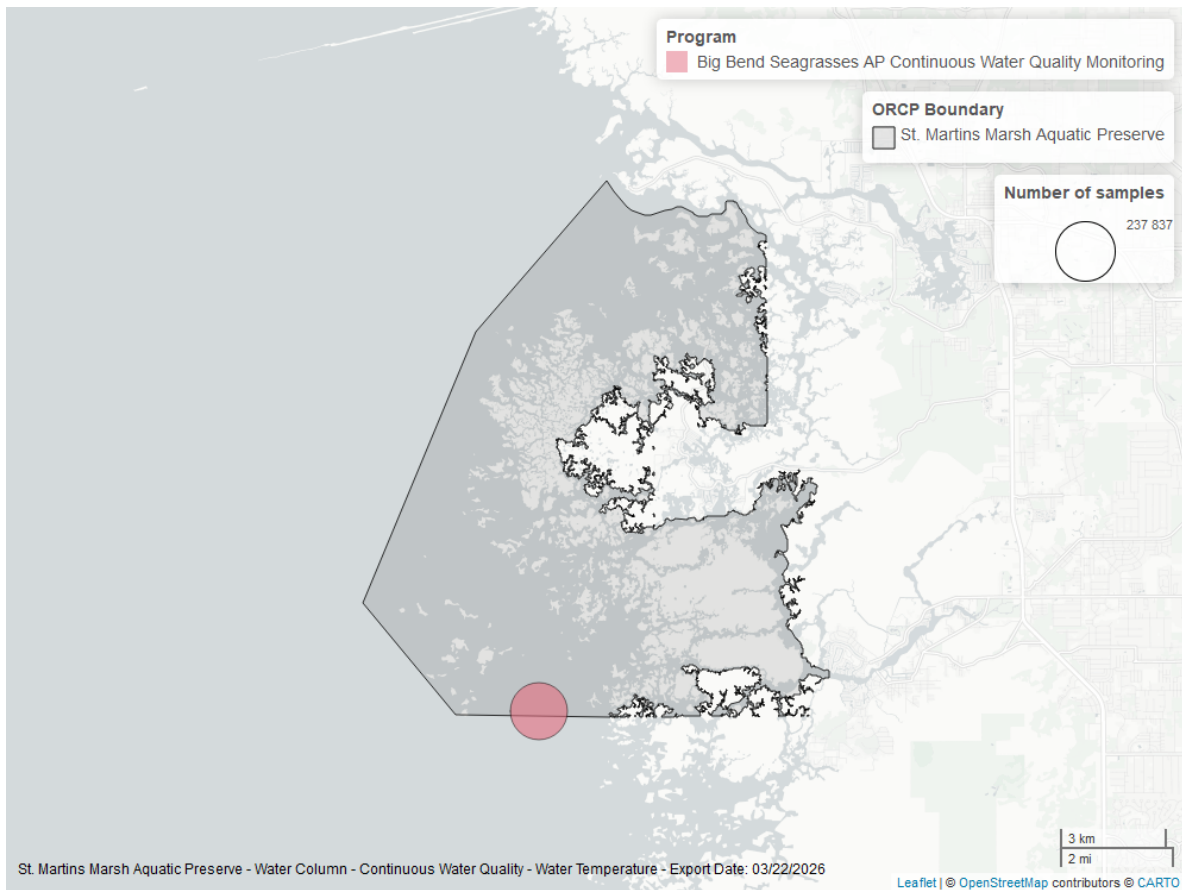


Figure 18: Map showing location of water temperature continuous water quality sampling locations within the boundaries of *St. Martins Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## Water Temperature - Discrete

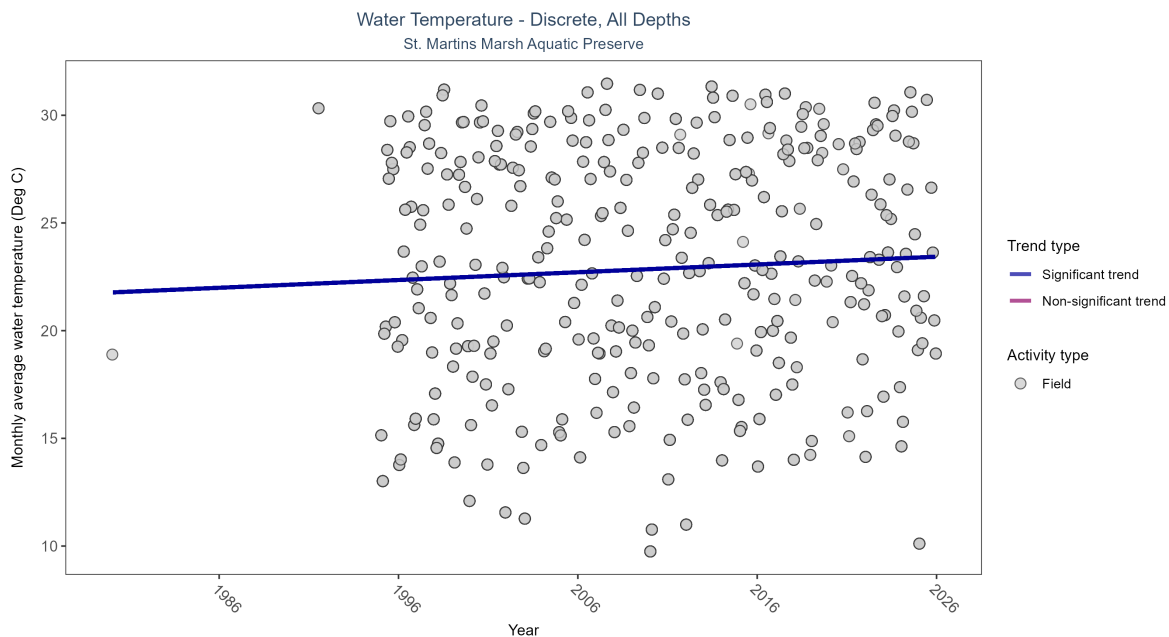


Figure 19: Scatter plot of monthly average water temperature over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only water temperature measurements taken in the field (circles) are included in the plot.

Table 10: Seasonal Kendall-Tau Results for - Water Temperature

| Activity Type | Statistical Trend              | Sample Count | Years with Data | Period of Record | Median Result Value | Tau     | Sen Intercept | Sen Slope | P     |
|---------------|--------------------------------|--------------|-----------------|------------------|---------------------|---------|---------------|-----------|-------|
| Field         | Significantly increasing trend | 8501         | 33              | 1980 - 2025      | 22.6                | 0.11732 | 21.7742       | 0.03605   | 0.003 |

Monthly average water temperature increased by 0.04°C per year.



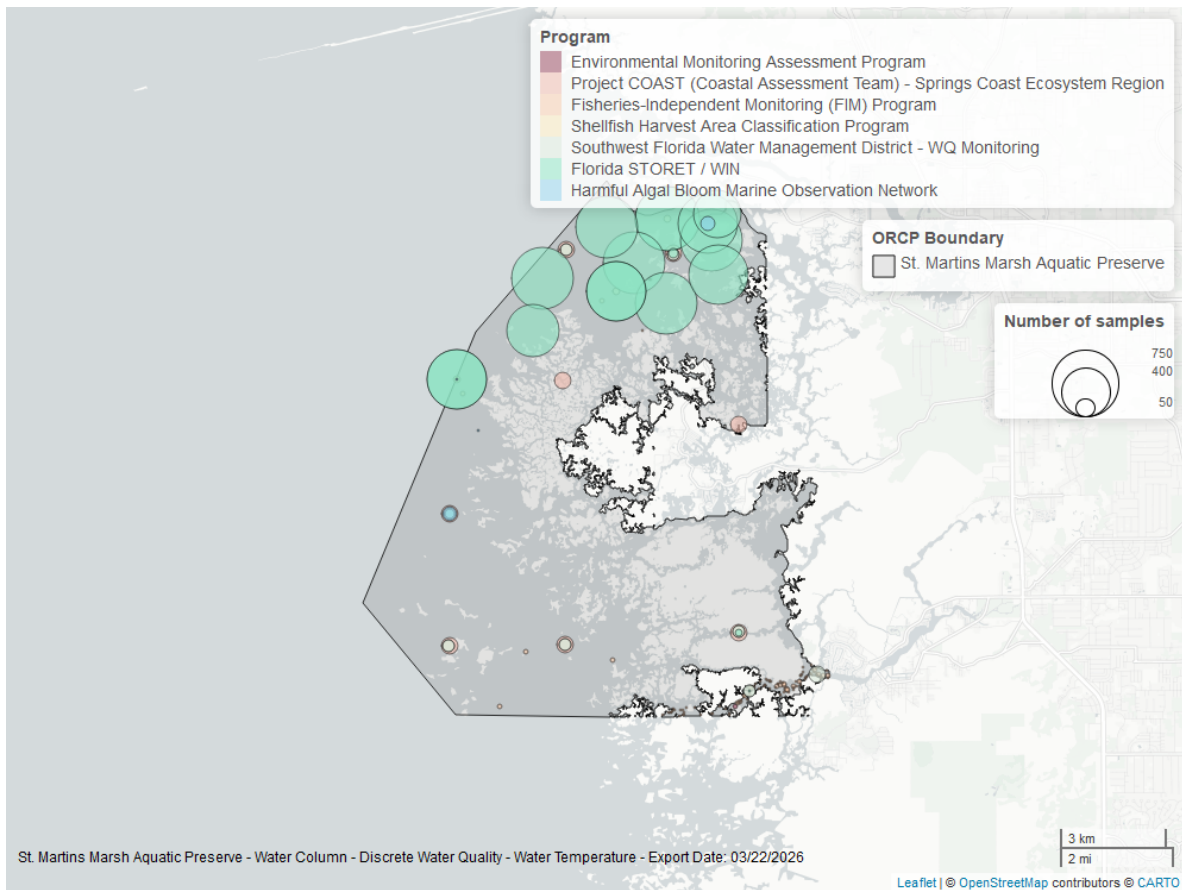


Figure 20: Map showing location of discrete water quality sampling locations within the boundaries of *St. Martins Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## pH - Continuous

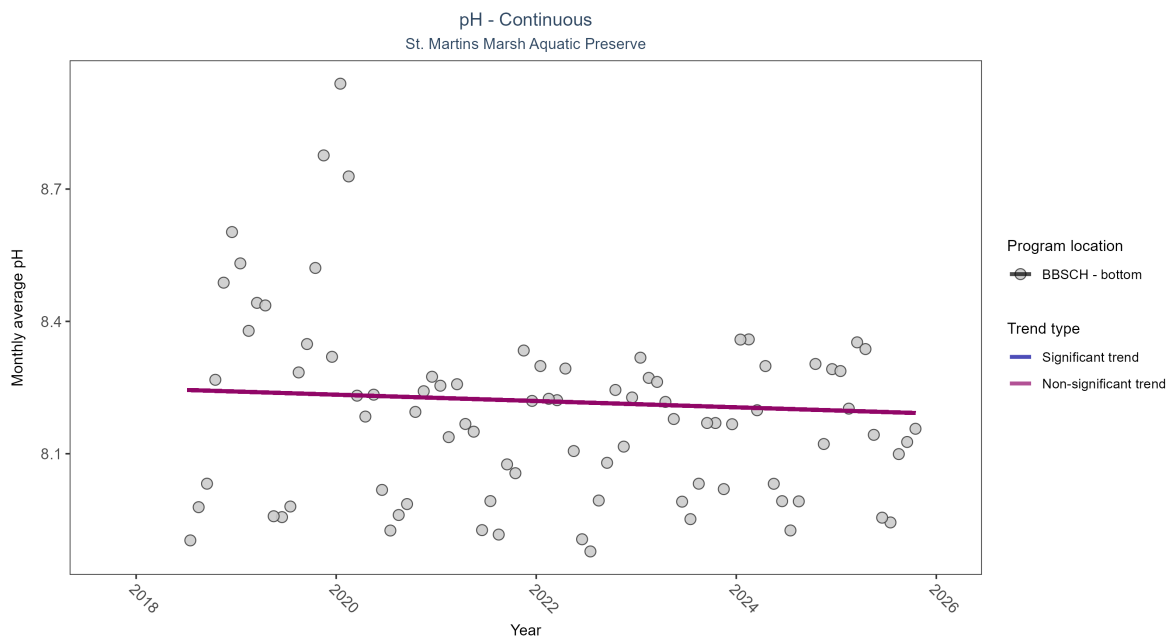


Figure 21: Scatter plot of monthly average pH over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 11: Seasonal Kendall-Tau Results - pH

| Program Location | Statistical Trend    | Sample Count | Years with Data | Period of Record | Median Result Value | Tau  | Sen Intercept | Sen Slope | P      |
|------------------|----------------------|--------------|-----------------|------------------|---------------------|------|---------------|-----------|--------|
| BBSCH            | No significant trend | 229419       | 8               | 2018 - 2025      | 8.2                 | -0.1 | 8.25          | -0.01     | 0.2865 |

No detectable change in monthly average pH was observed at one location.

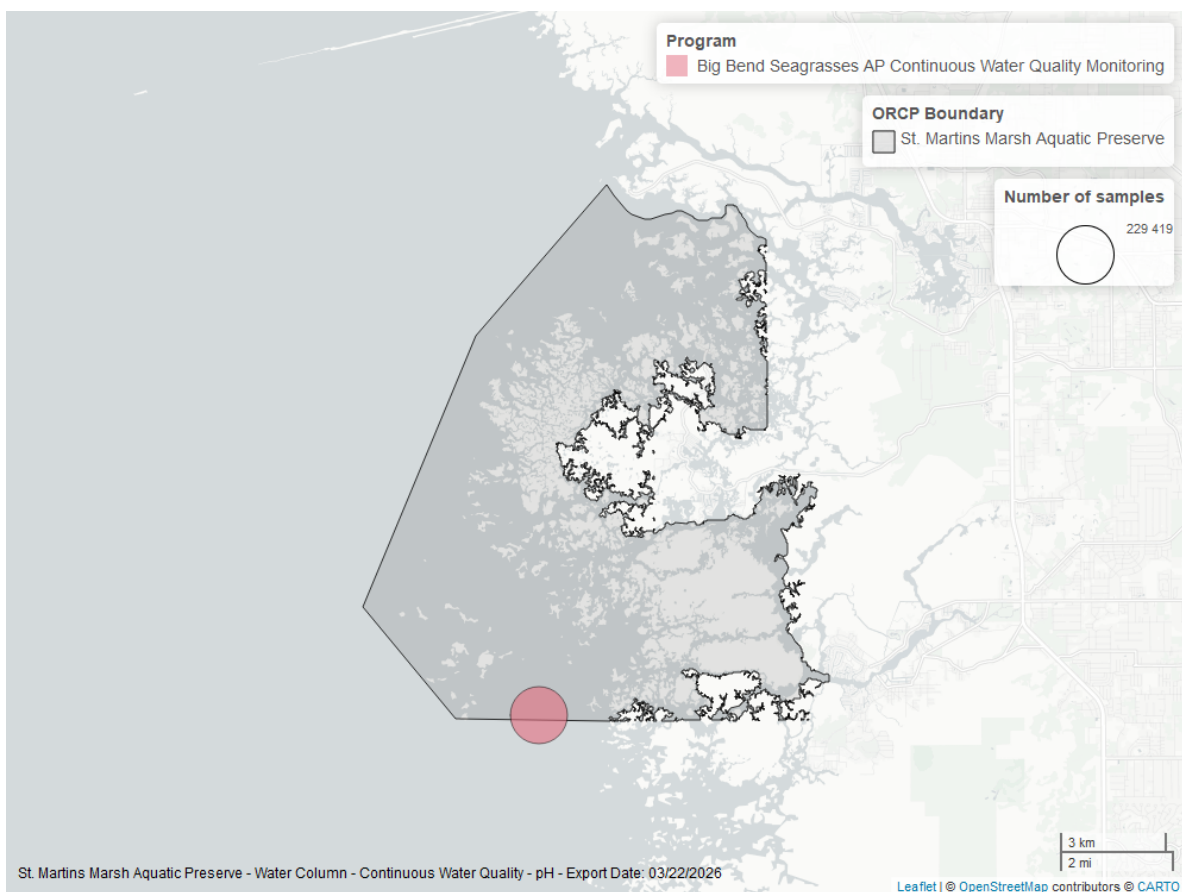


Figure 22: Map showing location of pH continuous water quality sampling locations within the boundaries of *St. Martins Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## pH - Discrete

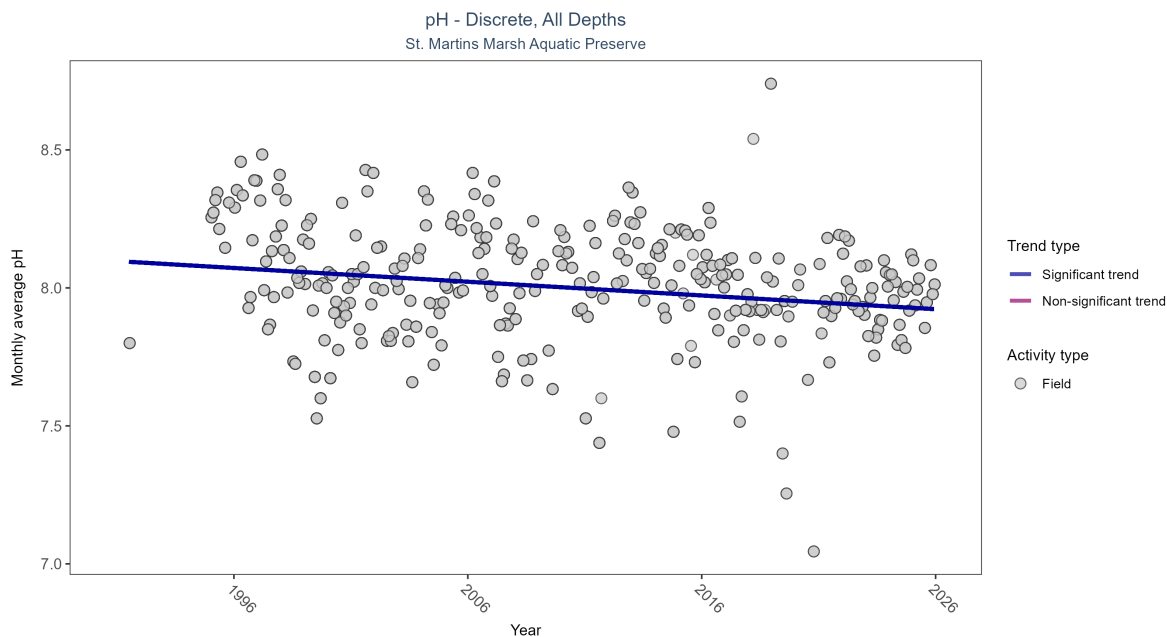


Figure 23: Scatter plot of monthly average pH over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only pH values measured in the field (circles) are included in the plot.

Table 12: Seasonal Kendall-Tau Results for - pH

| Activity Type | Statistical Trend              | Sample Count | Years with Data | Period of Record | Median Result Value | Tau      | Sen Intercept | Sen Slope | P |
|---------------|--------------------------------|--------------|-----------------|------------------|---------------------|----------|---------------|-----------|---|
| Field         | Significantly decreasing trend | 4421         | 32              | 1991 - 2025      | 8.09                | -0.16693 | 8.09704       | -0.00498  | 0 |

Monthly average pH decreased by less than 0.01 pH units per year.

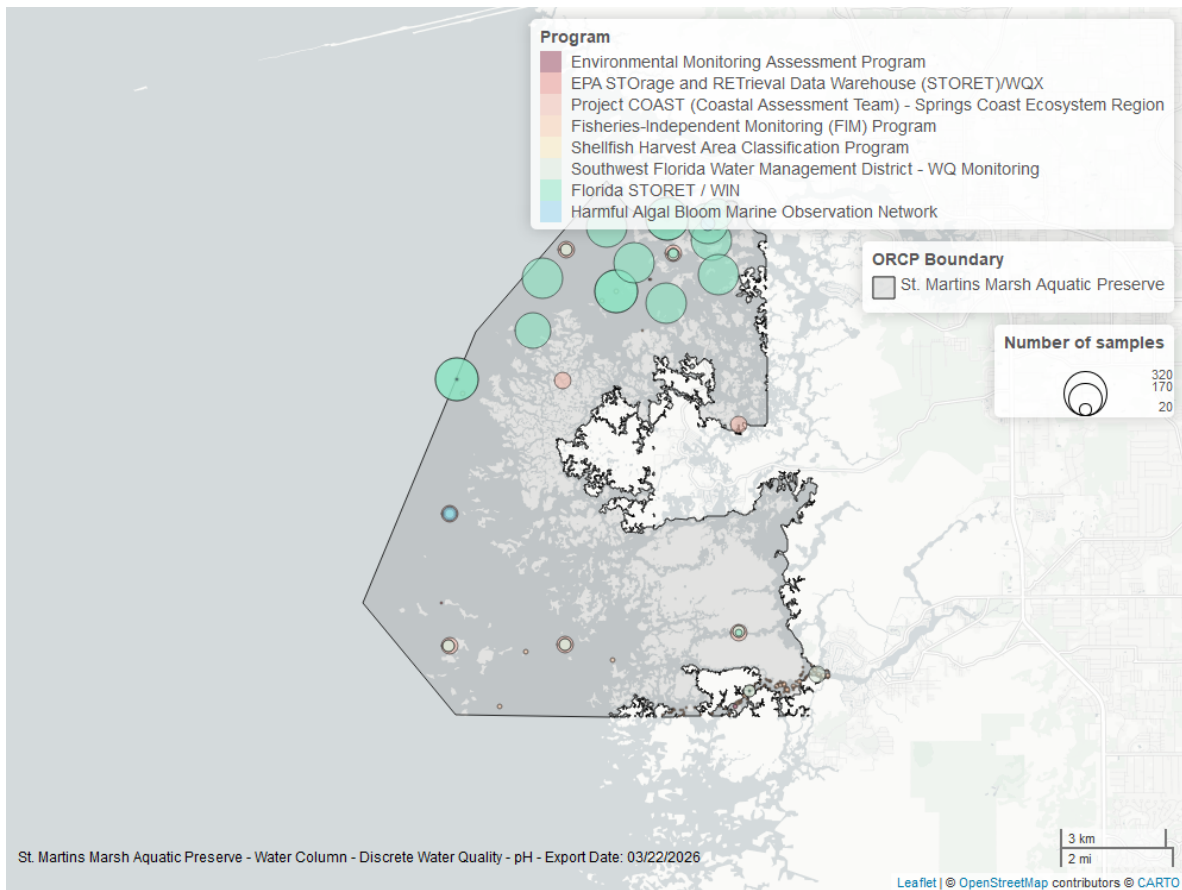


Figure 24: Map showing location of discrete water quality sampling locations within the boundaries of *St. Martins Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## Specific Conductivity - Continuous

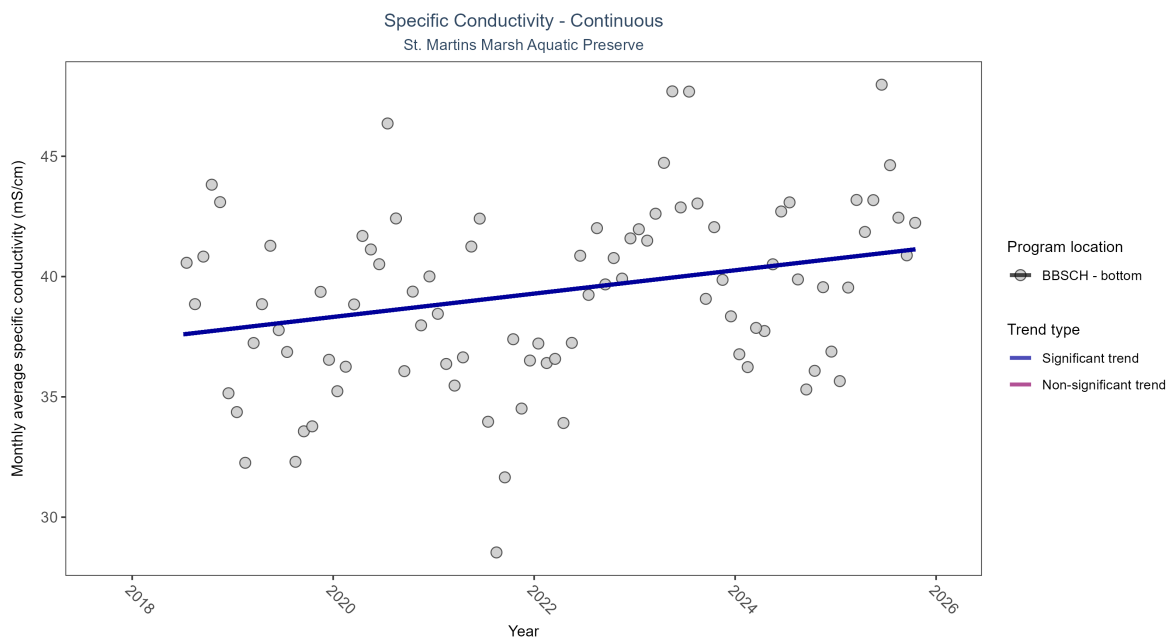


Table 13: Seasonal Kendall-Tau Results - Specific Conductivity

| Program Location | Statistical Trend              | Sample Count | Years with Data | Period of Record | Median Result Value | Tau  | Sen Intercept | Sen Slope | P      |
|------------------|--------------------------------|--------------|-----------------|------------------|---------------------|------|---------------|-----------|--------|
| BBSCH            | Significantly increasing trend | 234051       | 8               | 2018 - 2025      | 39.55               | 0.27 | 37.35         | 0.49      | 0.0025 |

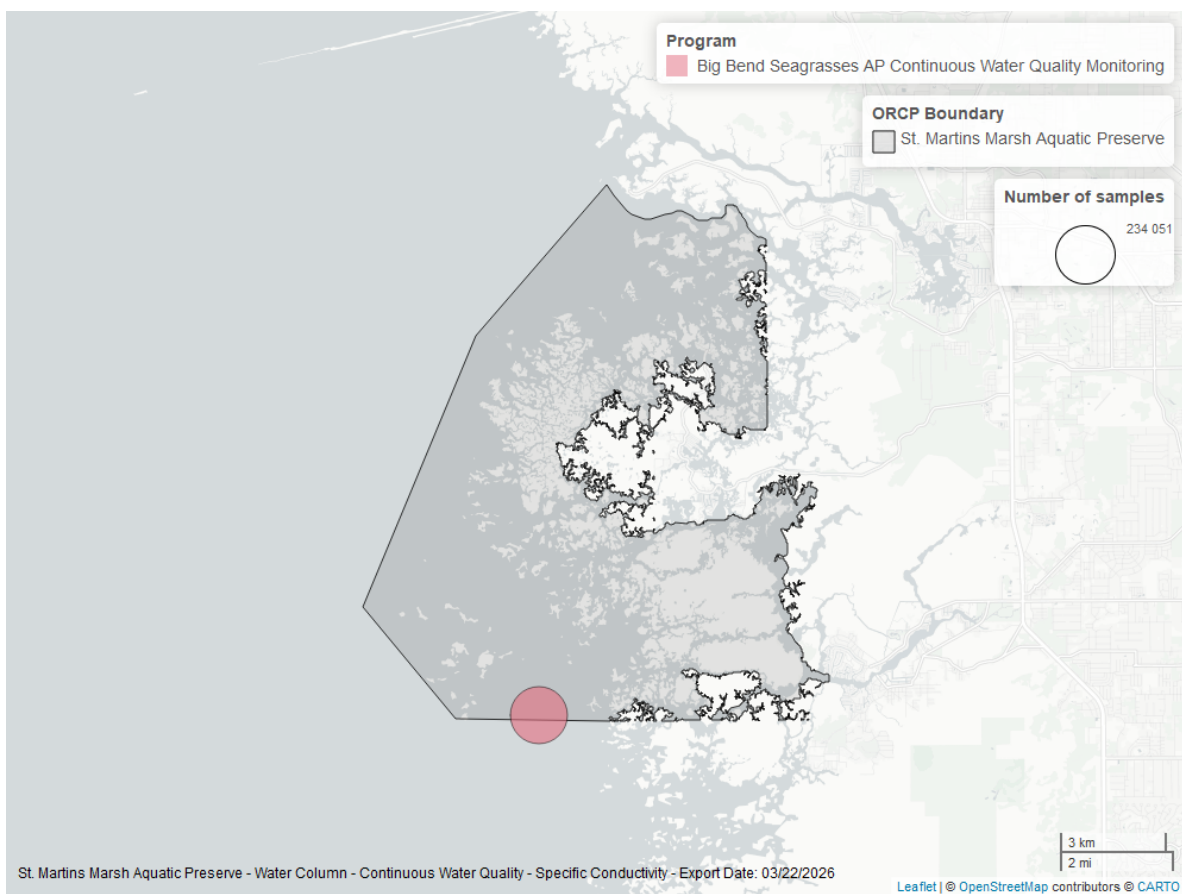


Figure 25: Map showing location of specific conductivity continuous water quality sampling locations within the boundaries of *St. Martins Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## Water Clarity

### Turbidity - Continuous

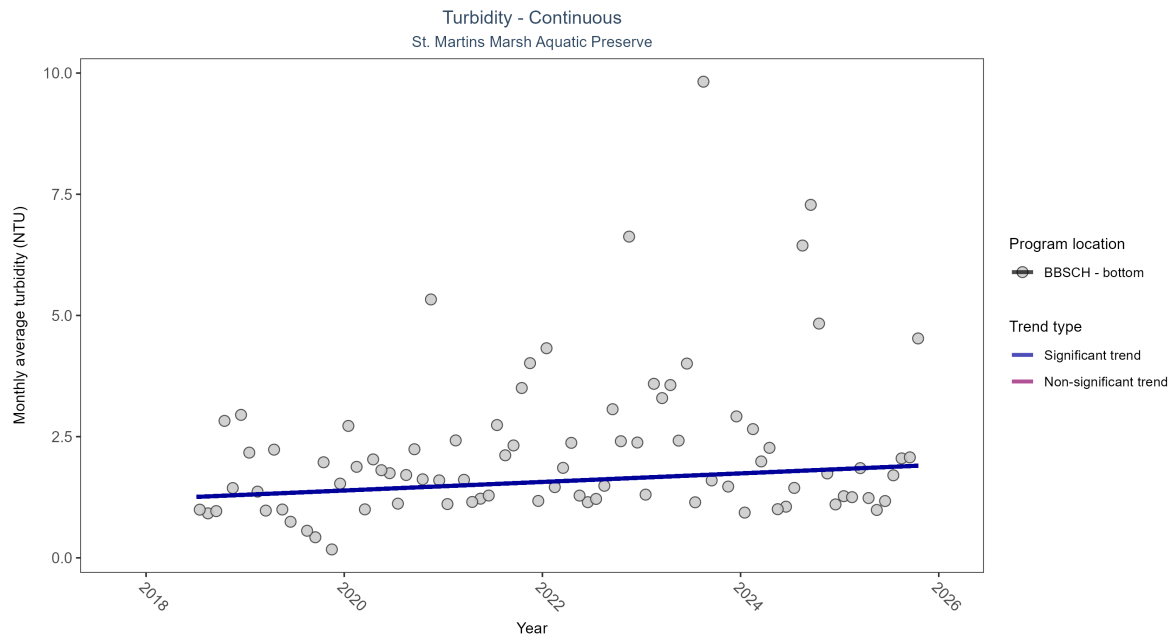


Figure 26: Scatter plot of monthly average turbidity over time at continuously monitored program locations. Each location is analyzed separately, with significant (blue) or non-significant (magenta) trend lines shown for time series that included five or more years of observations.

Table 14: Seasonal Kendall-Tau Results - Turbidity

| Program Location | Statistical Trend              | Sample Count | Years with Data | Period of Record | Median Result Value | Tau  | Sen Intercept | Sen Slope | P     |
|------------------|--------------------------------|--------------|-----------------|------------------|---------------------|------|---------------|-----------|-------|
| BBSCH            | Significantly increasing trend | 204374       | 8               | 2018 - 2025      | 1                   | 0.19 | 1.21          | 0.09      | 0.027 |

At one program location, monthly average turbidity increased by 0.09 NTU per year.



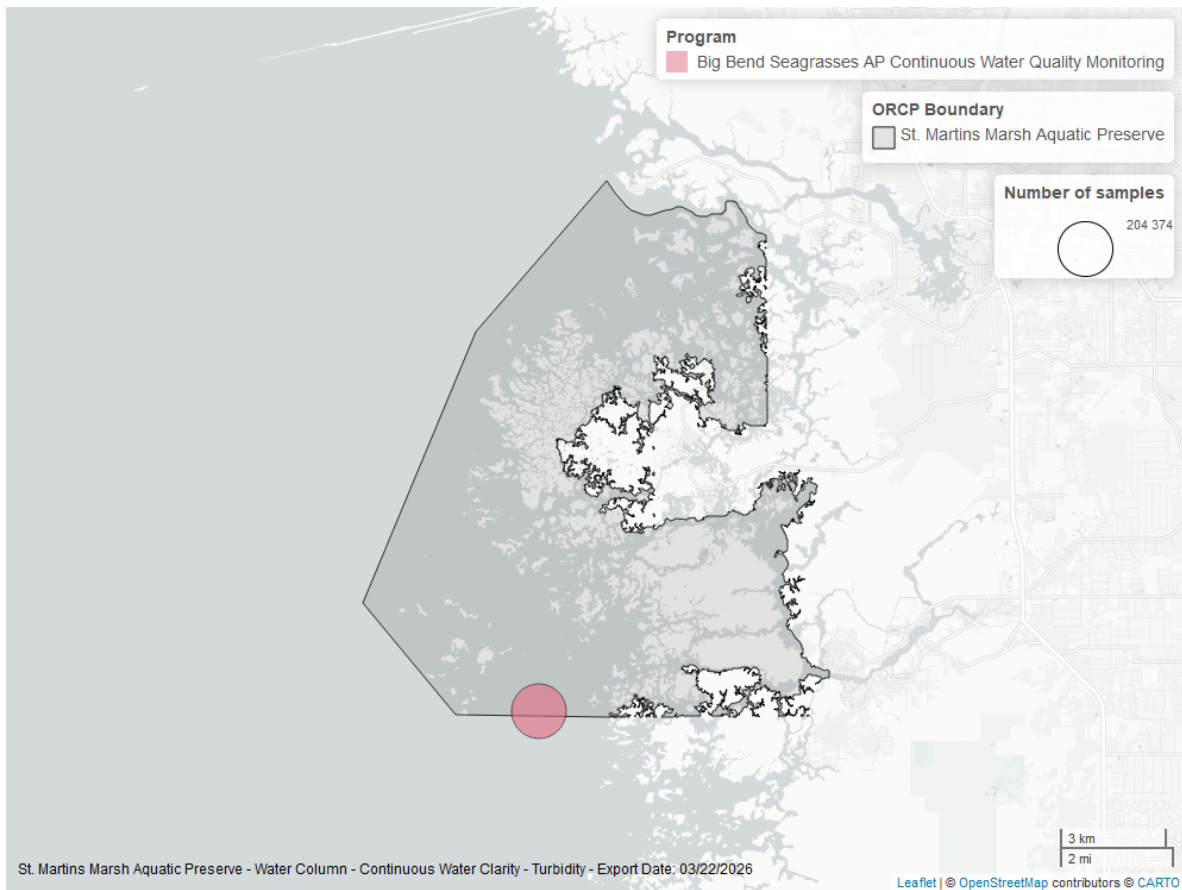


Figure 27: Map showing location of turbidity continuous water quality sampling locations within the boundaries of *St. Martins Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## Turbidity - Discrete

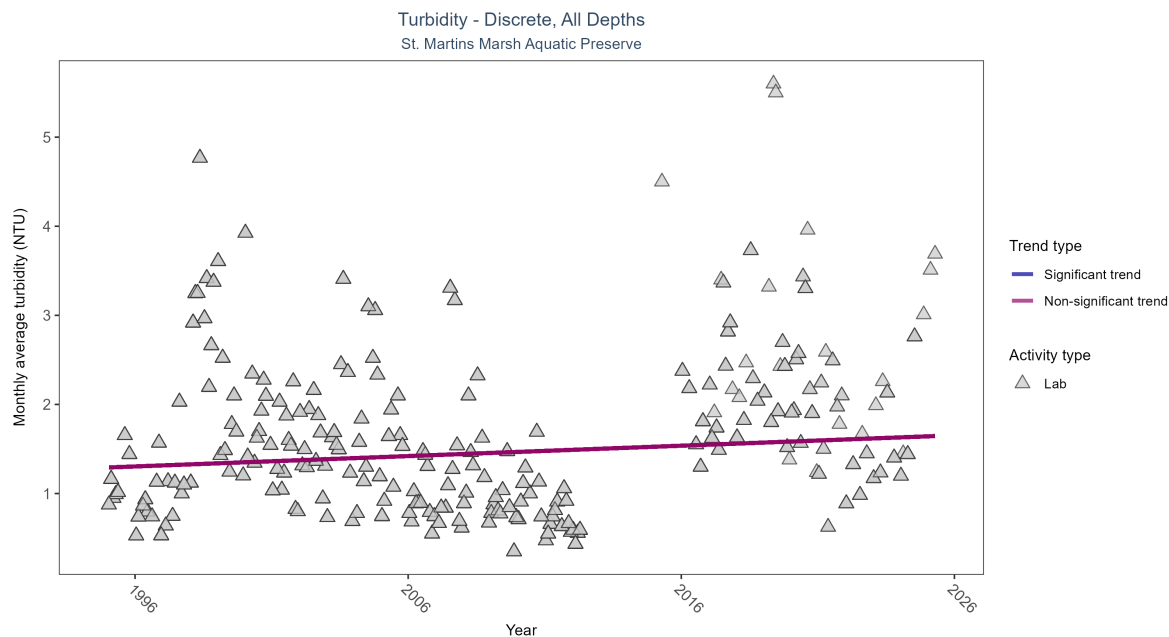


Figure 28: Scatter plot of monthly average turbidity over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only turbidity values measured in the laboratory (triangles) are included in the plot.

Table 15: Seasonal Kendall-Tau Results for - Turbidity

| Activity Type | Statistical Trend    | Sample Count | Years with Data | Period of Record | Median Result Value | Tau     | Sen Intercept | Sen Slope | P      |
|---------------|----------------------|--------------|-----------------|------------------|---------------------|---------|---------------|-----------|--------|
| Lab           | No significant trend | 3083         | 29              | 1995 - 2025      | 1.1                 | 0.07458 | 1.29219       | 0.01165   | 0.1002 |

Turbidity showed no detectable trend between 1995 and 2025.

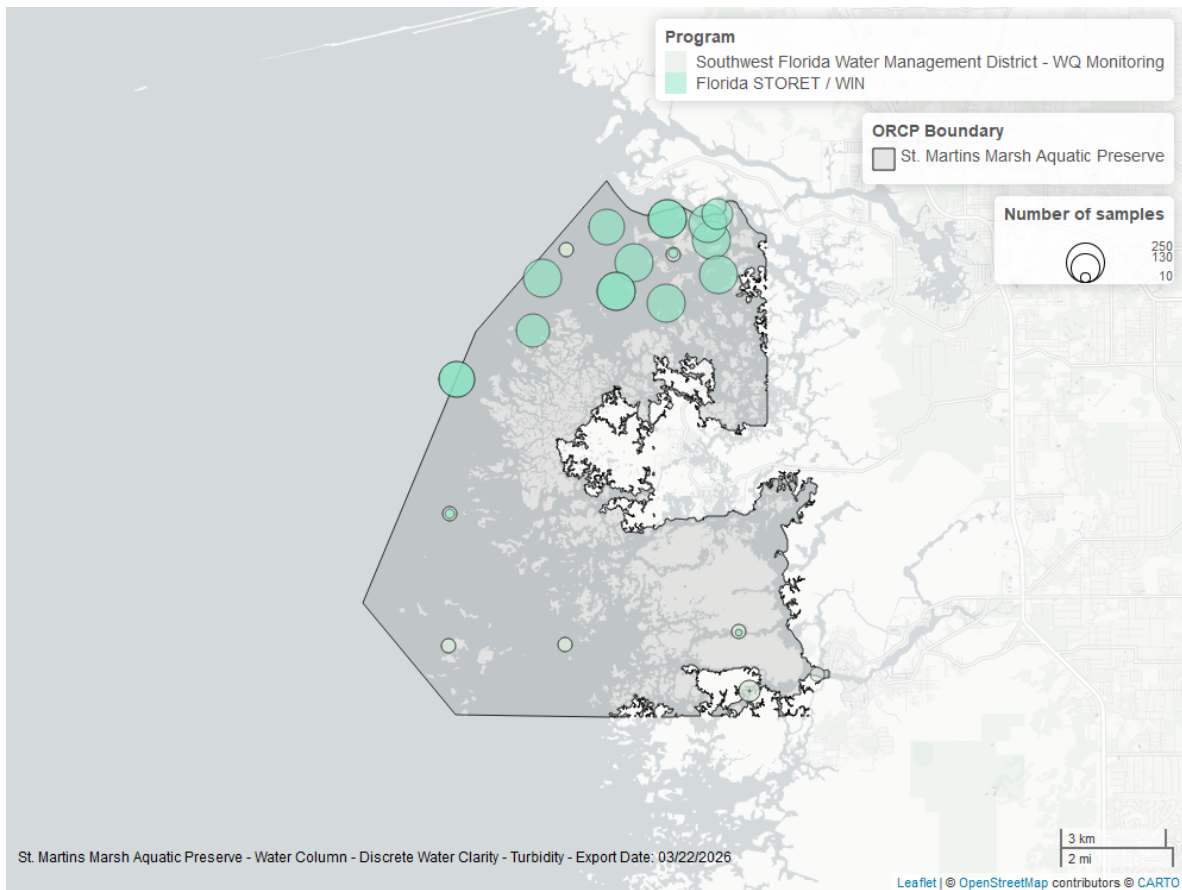


Figure 29: Map showing location of discrete water quality sampling locations within the boundaries of *St. Martins Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## Total Suspended Solids - Discrete

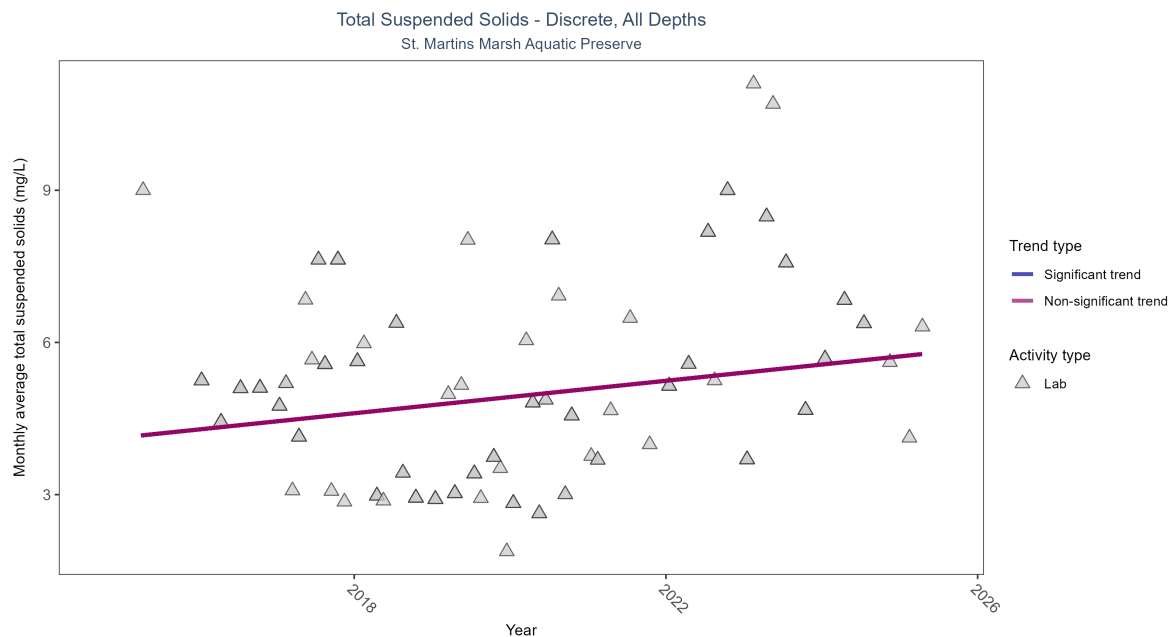


Figure 30: Scatter plot of monthly average total suspended solids (TSS) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only TSS values obtained from laboratory analyses (triangles) are included in the plot.

Table 16: Seasonal Kendall-Tau Results for - Total Suspended Solids

| Activity Type | Statistical Trend    | Sample Count | Years with Data | Period of Record | Median Result Value | Tau     | Sen Intercept | Sen Slope | P      |
|---------------|----------------------|--------------|-----------------|------------------|---------------------|---------|---------------|-----------|--------|
| Lab           | No significant trend | 256          | 11              | 2015 - 2025      | 4.63                | 0.14354 | 4.12333       | 0.16009   | 0.1693 |

Total suspended solids showed no detectable trend between 2015 and 2025.

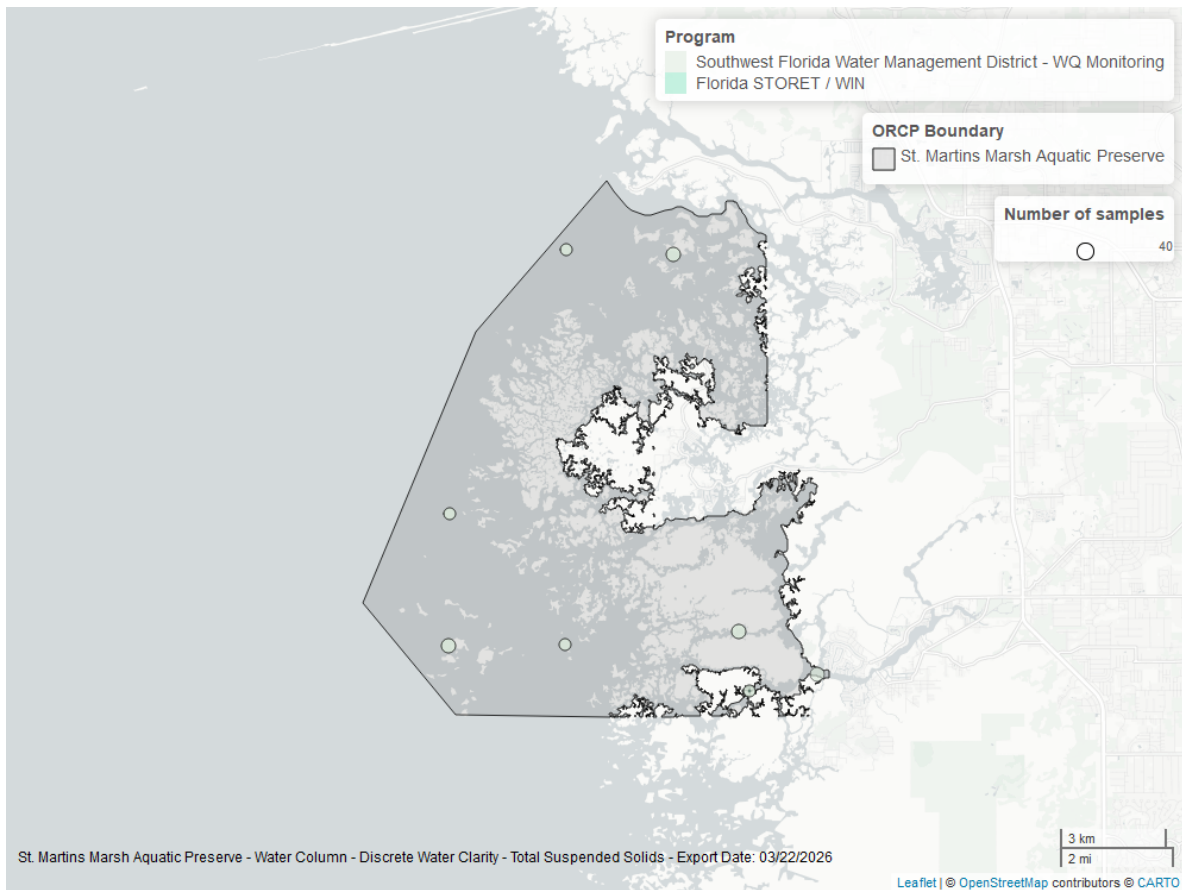


Figure 31: Map showing location of discrete water quality sampling locations within the boundaries of *St. Martins Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## Chlorophyll a, Uncorrected for Pheophytin - Discrete

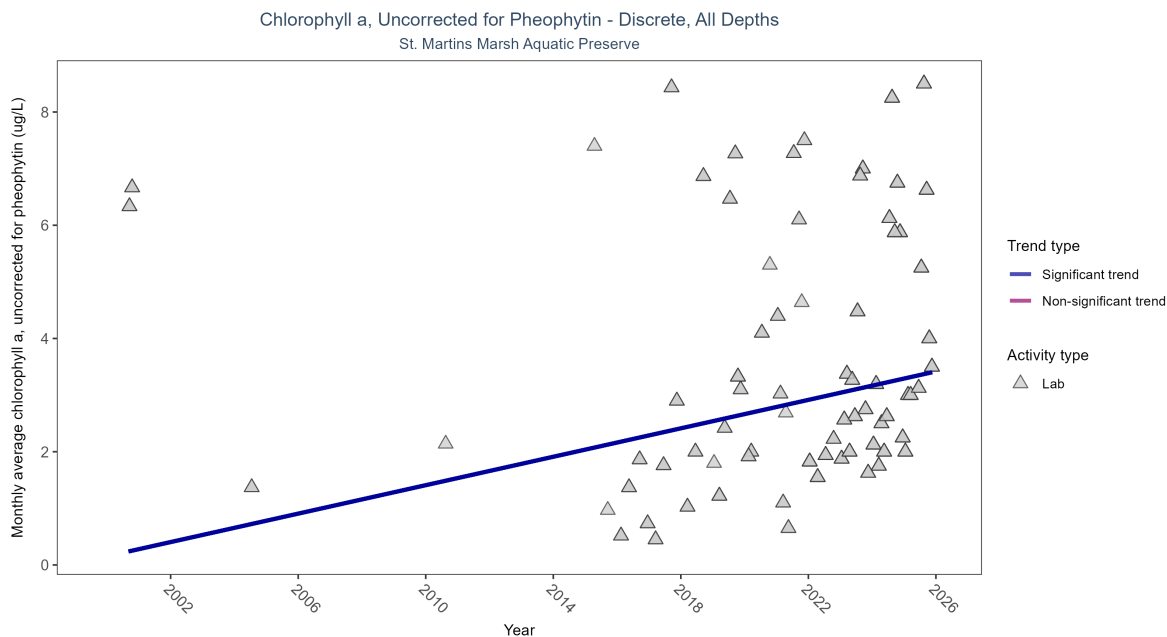


Figure 32: Scatter plot of monthly average levels of chlorophyll a, uncorrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 17: Seasonal Kendall-Tau Results for - Chlorophyll a, Uncorrected for Pheophytin

| Activity Type | Statistical Trend              | Sample Count | Years with Data | Period of Record | Median Result Value | Tau     | Sen Intercept | Sen Slope | P      |
|---------------|--------------------------------|--------------|-----------------|------------------|---------------------|---------|---------------|-----------|--------|
| Lab           | Significantly increasing trend | 384          | 14              | 2000 - 2025      | 2.225               | 0.31679 | 0.15318       | 0.12556   | 0.0107 |

Monthly average chlorophyll a, uncorrected for pheophytin, increased by 0.13 µg/L per year, indicating a decrease in water clarity.

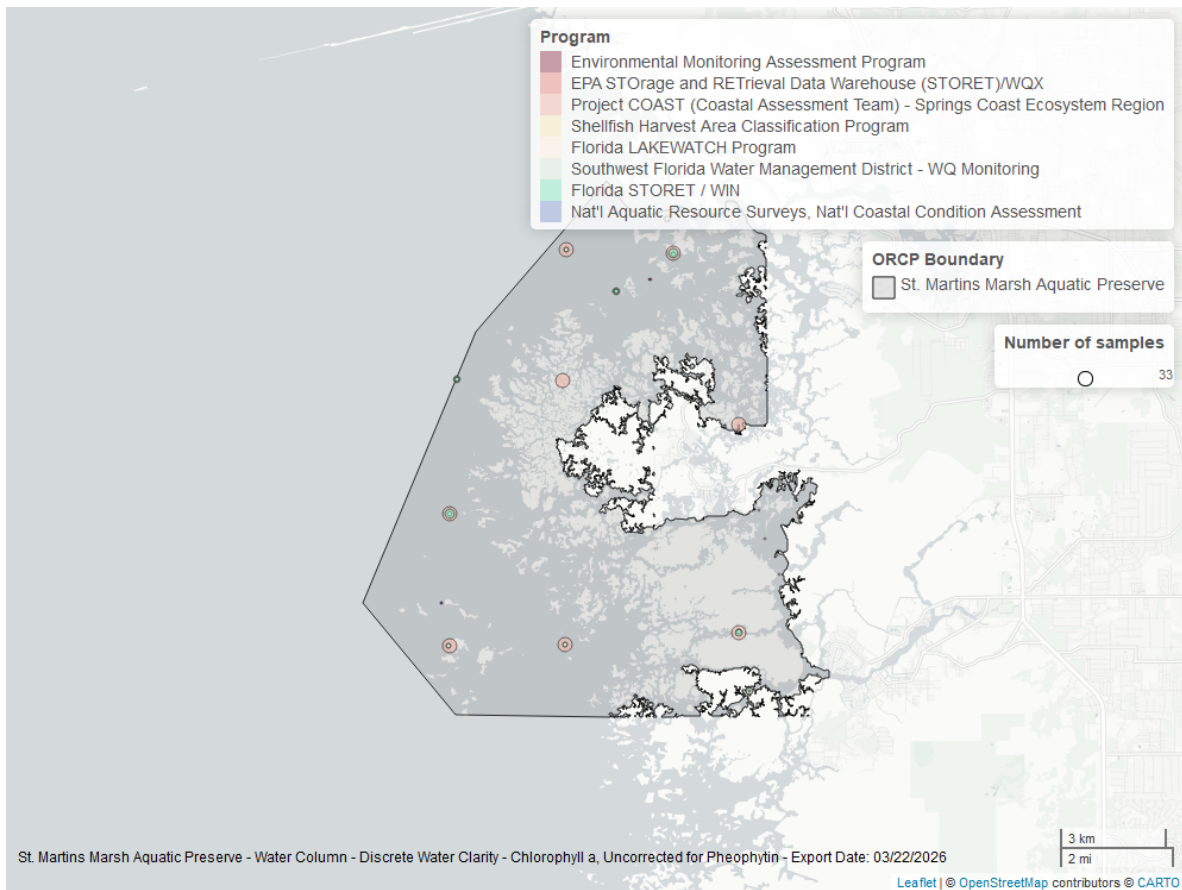


Figure 33: Map showing location of discrete water quality sampling locations within the boundaries of *St. Martins Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## Chlorophyll a, Corrected for Pheophytin - Discrete

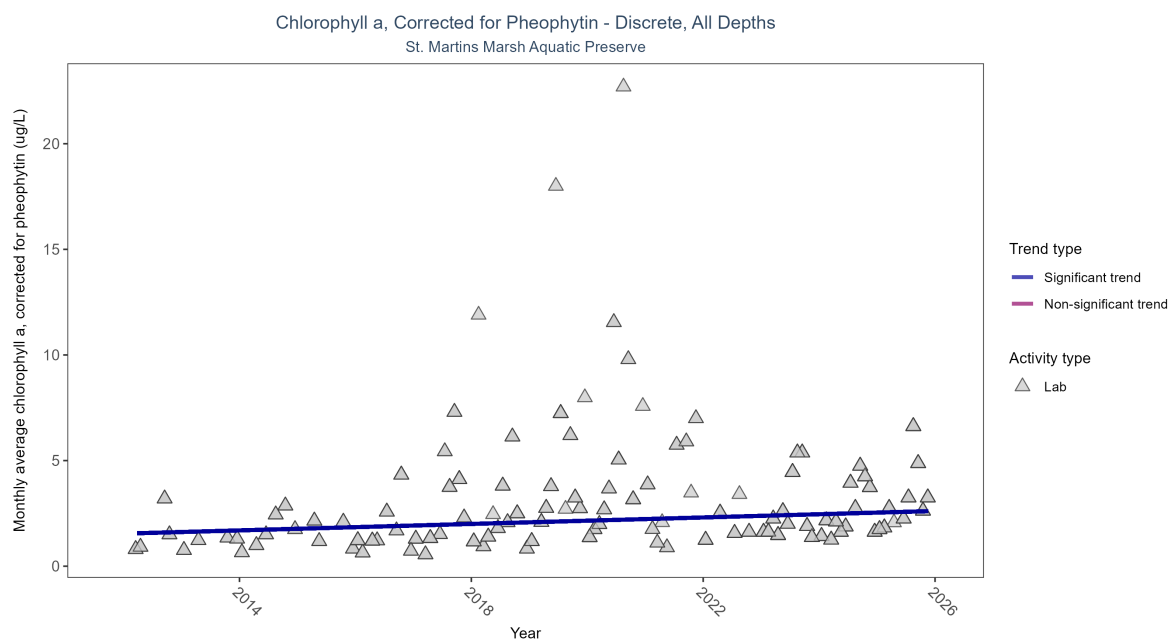


Figure 34: Scatter plot of monthly average levels of chlorophyll a, corrected for pheophytin, over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed chlorophyll a (triangles) is included in the plot.

Table 18: Seasonal Kendall-Tau Results for - Chlorophyll a, Corrected for Pheophytin

| Activity Type | Statistical Trend              | Sample Count | Years with Data | Period of Record | Median Result Value | Tau     | Sen Intercept | Sen Slope | P      |
|---------------|--------------------------------|--------------|-----------------|------------------|---------------------|---------|---------------|-----------|--------|
| Lab           | Significantly increasing trend | 635          | 14              | 2012 - 2025      | 1.8                 | 0.27736 | 1.54162       | 0.07657   | 0.0001 |

Monthly average chlorophyll a, corrected for pheophytin, increased by 0.08  $\mu\text{g/L}$  per year, indicating a decrease in water clarity.



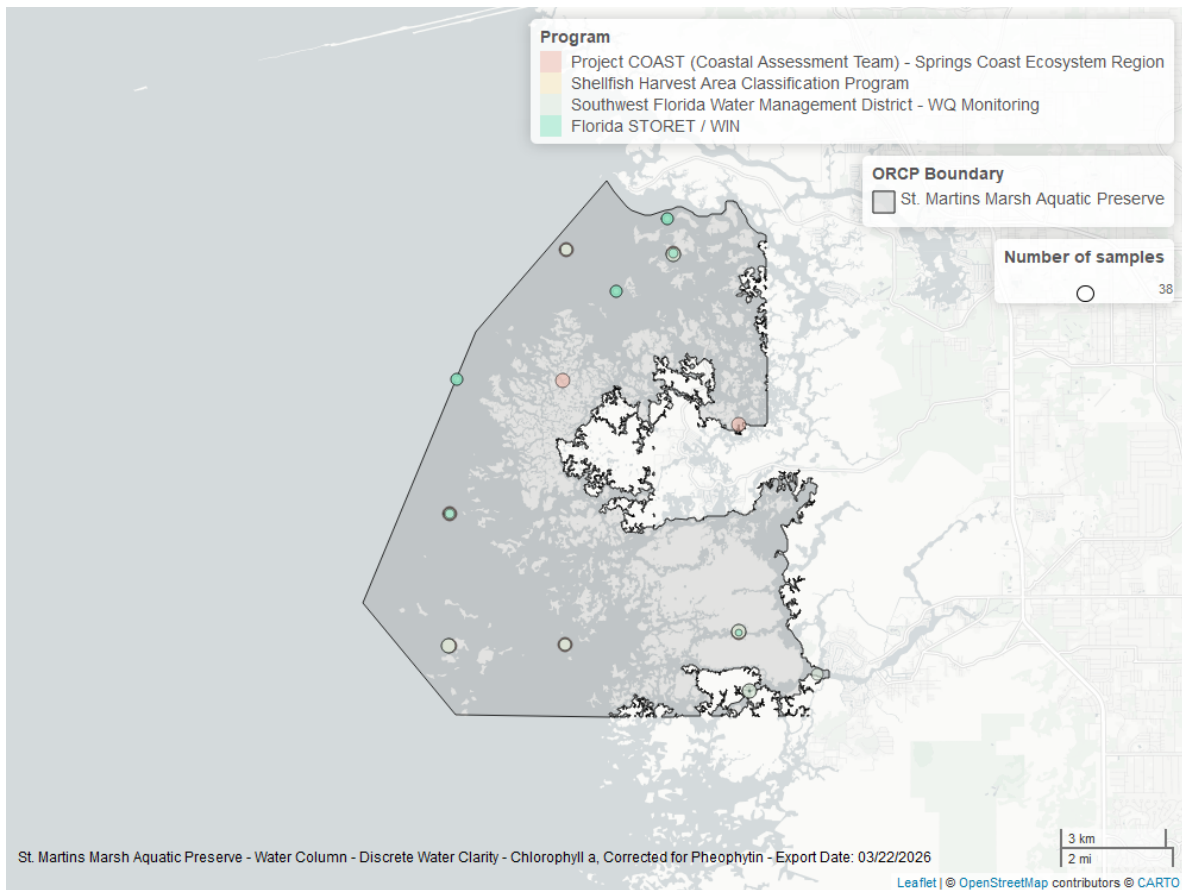


Figure 35: Map showing location of discrete water quality sampling locations within the boundaries of *St. Martins Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## Secchi Depth - Discrete

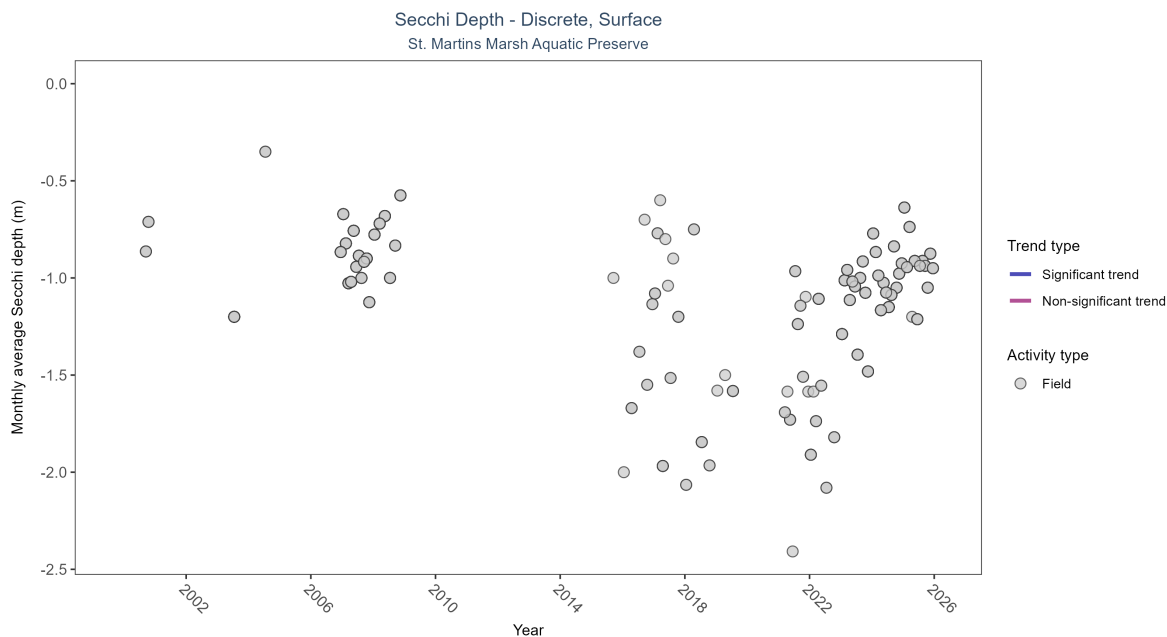


Figure 36: Scatter plot of monthly average Secchi depth over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Secchi depth is only measured in the field (circles).

Table 19: Seasonal Kendall-Tau Results for - Secchi Depth

| Activity Type | Statistical Trend              | Sample Count | Years with Data | Period of Record | Median Result Value | Tau     | Sen Intercept | Sen Slope | P      |
|---------------|--------------------------------|--------------|-----------------|------------------|---------------------|---------|---------------|-----------|--------|
| Field         | Significantly decreasing trend | 815          | 17              | 1991 - 2025      | -0.91441            | -0.1585 | -0.79544      | -0.00636  | 0.0311 |

Monthly average Secchi depth became deeper by 0.01 m per year, indicating an increase in water clarity.

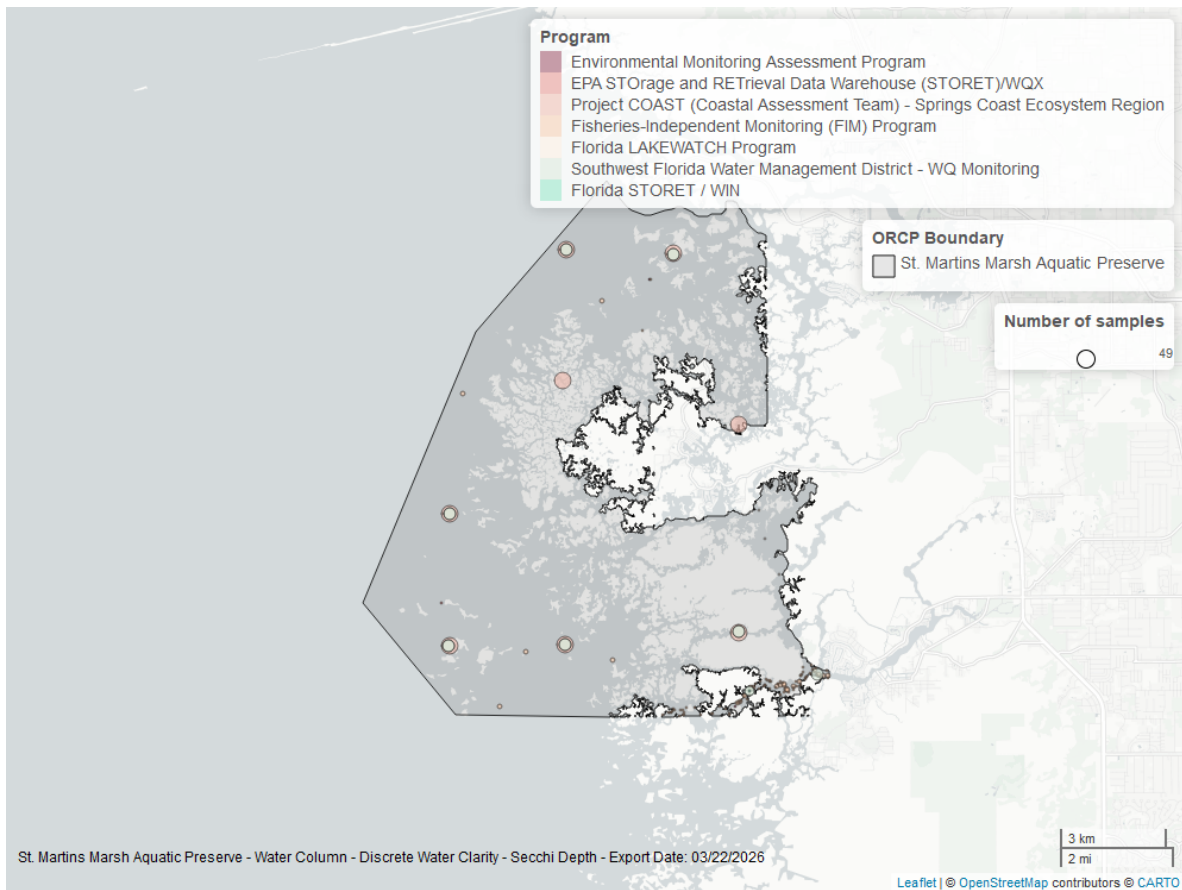


Figure 37: Map showing location of discrete water quality sampling locations within the boundaries of *St. Martins Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.

## Colored Dissolved Organic Matter - Discrete

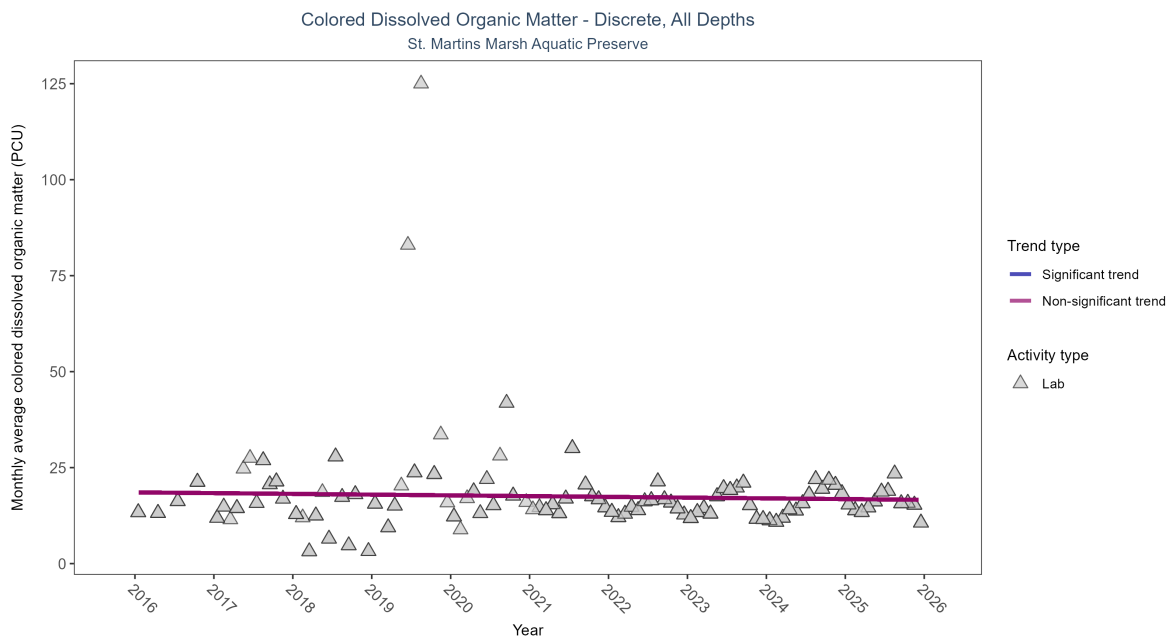


Figure 38: Scatter plot of monthly average colored dissolved organic matter (CDOM) over time. If the time series included ten or more years of discrete observations, a significant (blue) or non-significant (magenta) trend line is also shown. Only laboratory-analyzed CDOM (triangles) is included in the plot.

Table 20: Seasonal Kendall-Tau Results for - Colored Dissolved Organic Matter

| Activity Type | Statistical Trend    | Sample Count | Years with Data | Period of Record | Median Result Value | Tau     | Sen Intercept | Sen Slope | P      |
|---------------|----------------------|--------------|-----------------|------------------|---------------------|---------|---------------|-----------|--------|
| Lab           | No significant trend | 731          | 10              | 2016 - 2025      | 13.9                | -0.1162 | 18.54988      | -0.19459  | 0.1619 |

Colored dissolved organic matter showed no detectable trend between 2016 and 2025.

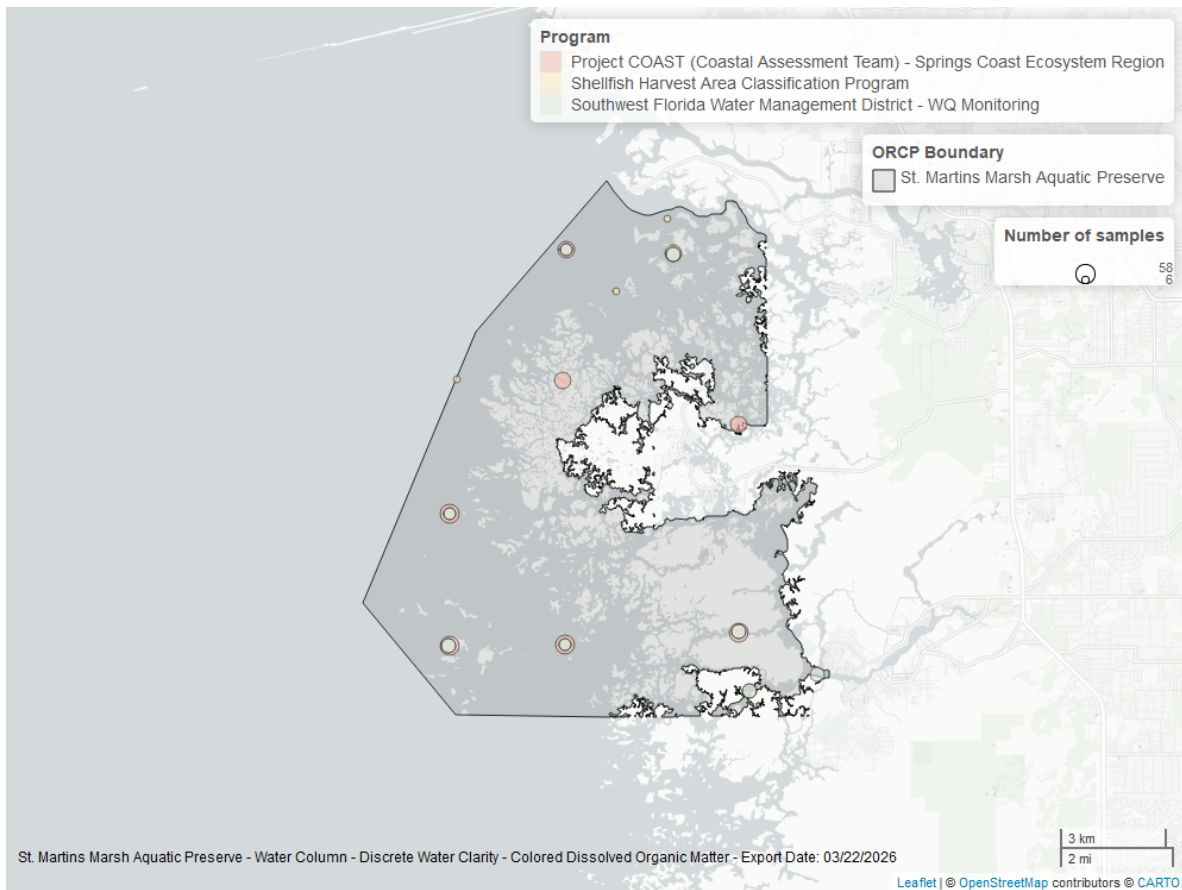


Figure 39: Map showing location of discrete water quality sampling locations within the boundaries of *St. Martins Marsh Aquatic Preserve*. The bubble size on the maps above reflect the amount of data available at each sampling site.